



RESEARCH

Multi-Objective Optimization of a Three-Level Sustainable Food Supply Chain: Modeling the Impact of Government Subsidies

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Abstract This study develops an integrated optimization framework which supports the sustainable design of a food supply chain with three echelons: suppliers, a central manufacturer, and retailers. The model minimizes total cost and carbon emissions while simultaneously maximizing the share of products made with certified green processes, capturing economic, environmental, and social pillars of sustainability. Government policy is represented through two distinct incentives: a per-unit subsidy for green production and a per-use subsidy for alternative fuel vehicles, both directly reducing relevant costs in the decision space. For scalability, a tailored non-dominated sorting genetic algorithm II (NSGA-II) is developed and

benchmarked against the exact solution method. Computational experiments based on the data of a dairy products case study indicate that carefully calibrated policy incentives can cut the total system cost by more than 40% and reduce greenhouse gas emissions by around 25% while raising the share of green output to above 80%. The results also indicated a critical range of subsidy values that trigger rapid adoption of clean technologies and demonstrate diminishing marginal returns beyond that range. Comparative tests confirmed that the heuristic achieves solutions within 1% of proven Pareto fronts on moderate examples and maintains high solution quality with substantial time savings on larger problems. The study provides an integrated tool for researchers and decision-makers to align economic performance with environmental and social goals, and it offers actionable guidance on subsidy design for low-carbon resilient food supply chain networks.

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Introduction

The accelerating urgency of climate change and resource constraints has turned global attention to the sustainability of critical industries, including the food supply chain (Ahmad et al., 2017). Food systems claim a significant share of greenhouse gas emissions and environmental impacts, whose transformation is central to achieving societal climate and health goals (Sandberg et al., 2023). Globally, policymakers are rolling out optimal strategies

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and initiatives, such as subsidies for green technology and mandates on low-emission vehicles, to guide supply chain practices toward sustainability (Jin et al., 2024; Shi et al., 2024). In particular, the widespread adoption of alternative fuel vehicles (AFVs), including electric and hydrogen-powered trucks, is being promoted through financial incentives and regulatory targets, reshaping distribution networks across all tiers of the food sector (Ge & Chen, 2025; Singh & Pandey, 2024). These governmental strategies and measures intend to accelerate the transition from traditional fossil-fuel-based logistics to eco-friendly transportation modes, as evidenced by various national programs that subsidize AFVs for freight delivery (Axsen & Pickrell-Barr, 2024). The combined pressure of these policy interventions and evolving consumer preferences for sustainability highlights a profound practical and theoretical motivation to model how subsidies and AFV adoption interact in supply chain decision-making (Bae et al., 2024). Effective implementation of green supply chain practices relies on drivers such as environmental regulations and top-management commitment, which can be shaped by well-targeted subsidy schemes to accelerate uptake across all echelons of a network. Designing green supply chain contracts that distribute environmental responsibility among partners aligns profitability with ecological objectives, indicating the importance of differentiated incentive structures (Das et al., 2022a). Coordinating agri-food networks through revenue-sharing contracts which engage retailers in green marketing efforts can simultaneously boost consumer adoption and environmental performance, offering a concrete pathway for policy firm collaboration (Das et al., 2022b, 2024).

Beyond these practical drivers, academic interest has surged in developing multi-objective optimization (MOO) models which capture economic and environmental dimensions (He et al., 2024; Jin et al., 2024). In the production and distribution of food, these economic and ecological perspectives are especially important due to the industry's large environmental footprint and its significance in feeding populations (Rezakhanlou & Mirzapour Al-e-Hashem, 2025). Recent research emphasizes that supply chain strategies must balance profitability with carbon emission reductions and, in some cases, social welfare objectives, necessitating advanced modeling techniques (Chen et al., 2024; Wang, 2024). For example, Jin et al. (2024) demonstrate that government subsidies can significantly influence greening investments and social welfare outcomes of manufacturers in production systems, while He et al. (2024) show that dual-subsidy schemes may optimize emission abatement in low-carbon supply chains under information asymmetry. These studies highlight the theoretical need to integrate policy instruments into supply chain models. However, the literature treats government

incentives in isolation or focuses on simpler one- or two-tier chain settings (Guo et al., 2024; Li et al., 2022). In contrast, food supply chains involve multiple echelons, such as suppliers, manufacturers, and retailers, and reveal dynamics such as perishability and heterogeneity of products, which complicate the implementation of green technologies (Panigrahy & Panda, 2025). Indeed, subsidizing green practices can change pricing, investment, and marketing decisions in complex ways (Shi et al., 2022; Wu et al., 2023). The introduction of AFVs into food logistics creates further challenges and opportunities that have only recently begun to be explored. AFVs can result in large reductions in per-unit transportation emissions, while they impose higher capital and operational costs that need to be offset by economies of scale or subsidies (Wanniarachchi et al., 2023). It is noteworthy to mention that government support for AFVs, such as purchase rebates or tax exemptions, can accelerate their deployment, but may also generate unintended trade-offs between market share and environmental performance (Cantillo et al., 2022).

The literature indicates that stakeholder engagement can become an order-winning factor for sustainable food companies (Cosa & Torelli, 2024; D'Adamo, 2023). Through integrating farmers, processors, retailers and consumers into joint planning sessions and co-creation projects, firms not only strengthen legitimacy but also set themselves apart in contract awards (Pullman & Wu, 2021; Siems et al., 2022). Evidence shows that companies with structured multi-party governance achieve higher rates of contract renewal and bounce back more quickly when market conditions change (Yang et al., 2022). Concurrently, a flexible management approach supports sustainability by allowing supply networks to adjust environmental and equity priorities on the fly, reroute deliveries during disruptions and tweak inventory levels to account for perishability (Ivanov, 2022; Mokline & Ben Abdallah, 2022; Sarker et al., 2021). These adjustments help maintain service performance and emission goals even under volatility (Drent et al., 2023). Incorporating these engagement and agility practices into a multi-objective model shows how firms turn policy incentives and AFV adoption into lasting competitive advantages.

In spite of these insights, the literature lacks a comprehensive framework that simultaneously considers the financial incentives for green technologies, the multi-echelon supply chain structure of food products, and the multiple objectives of stakeholders. Existing green supply chain optimization studies focus on single objectives, such as minimizing cost or emissions or on bilateral supply chain coordination, and they seldom incorporate both government subsidies and technological shifts to AFVs in one model.

This gap is critical since government subsidies and AFVs are inextricably linked in practice: subsidy policies are explicitly designed to make AFVs cost-competitive, and widespread AFV adoption can only be achieved under the right policy mix. Further, there are no studies modeled a three-level food supply chain with embedded subsidy mechanisms for AFVs, evaluating how these interventions affect the chain's economic, environmental, and social performance.

To fill this gap, this paper develops a novel MOO model for a three-tier sustainable food supply chain which incorporates government subsidies to promote AFV adoption. In contrast to prior works, our model enables decision-makers at each echelon to invest in AFVs under a government subsidy structure, while simultaneously pursuing multiple objectives such as maximizing profit, minimizing carbon emissions, and enhancing social welfare. The main contributions of this study are summarized as follows:

- Formulating a three-level supply chain model (suppliers–manufacturers–retailers) that captures key green and social considerations in food logistics, including emissions, loss reduction, and welfare objectives.
- Integrating government subsidy mechanisms into the model, allowing for differentiated support (such as producer vs. consumer subsidies) that target the adoption and utilization of AFVs at each echelon.
- Conducting a multi-objective analysis using advanced optimization techniques to delineate trade-off frontiers and assess the effectiveness of subsidy policies in achieving triple-bottom-line outcomes.
- Providing managerial and policy insights by identifying conditions under which subsidies and AFVs create win-win scenarios for firms, consumers, and regulators, and by offering guidelines on how to allocate subsidies optimally across the supply chains.

This work advances the theoretical understanding of sustainable supply chain management through linking government subsidy policy with technology adoption in a complex food supply network. By doing so, it offers a unique contribution bridging gaps between the subsidization of green logistics and the optimization of food supply chains. Our findings would expectedly help scholars, decision-makers, and managers better appreciate the synergies and tensions between public incentives and private decision-making in building low-carbon food systems. This research, therefore, answers the following question:

How do government subsidy schemes for alternative fuel vehicles affect the economic, environmental, and social trade-offs in a multi-tier sustainable food supply chain, and under what conditions can these

subsidies most effectively enhance overall system performance?

The remainder of this paper is structured as follows: section “[Multi-echelon and perishability](#)” presents a comprehensive literature review of sustainable three-level food supply chains. Section “[Partial coverage of the triple bottom line objective space](#)” details the model formulation and describes its characteristics as a MOO problem. Section “Case Study, discussion, and sensitivity analysis” provides the numerical results, comparative metrics, and statistical analyses. Finally, section “Conclusion and future research directions” concludes the paper and offers recommendations for future research directions.

Literature Review

This section offers a comprehensive review of existing literature on food supply chain management (FSCM), with a focus on the critical aspects of sustainability and optimization approaches. The review starts by exploring the structure and challenges of food supply chains, followed by examining sustainability practices within the industry. Finally, it delves into optimization methods, including MOO, which contribute to the design and management of supply chain networks.

Food Supply Chain Management

The food industry is a diverse and complex sector that spans numerous stages from production to consumption, including agriculture, processing, distribution and retail (Augustin et al., 2016). Food supply chains (FSCs) therefore serve as integrated sequences of farming, processing, logistics and retail activities orchestrated through raw material, financial and information flows (Mishra et al., 2022). As global demand for food continues to rise, the industry faces increasing pressure to enhance efficiency, quality, and environmental sustainability while ensuring food security for both present and future generations (Prasanna et al., 2025). Addressing these multifaceted challenges necessitates robust FSCs that can adapt to evolving market and environmental conditions (Obonyo et al., 2025). Within this quest for resilience and sustainability, the configuration of the underlying supply network becomes a critical decision variable (Kiani Mavi et al., 2024). Recent studies on the network design of FSCs that incorporate dietary requirements suggest that the combined choice of supplier sourcing, processing location and transport mode reshapes total cost, nutritional adequacy, and greenhouse gas emissions in a complex relationship. Thus, strategic location and routing exert more influence on



sustainability than external demand shocks or price fluctuations (Tang et al., 2015).

Despite advances in network design, evidence on low-carbon logistics remains inconclusive. For instance, equity-weighted routing can reduce product loss and social disparity while holding public expenditure almost constant (Yontar, 2023). Conversely, shipment frequency models for generic consumer products report profit erosion once strict carbon caps are imposed, primarily because inventory-holding costs grow when shipment frequency falls (Gupta et al., 2025). Together, these contrasts suggest that the cost–emission frontier and trade-off are determined less by the carbon target itself and more by model scope, perishability assumptions, and the set of transport options (Tirkolaee & Aydin, 2022). Further, structural and geographical biases further constrain theoretical generalization. Over two-thirds of optimization case studies still rely on European dairy or North American meat data, while smallholder and public distribution networks in Africa, Latin America, and much of Asia remain largely unmodeled. The few investigations that have addressed emerging-economy contexts show different constraints, notably warehouse equity and last-mile access, yet these insights have not migrated into mainstream formulations (Shetty et al., 2024).

Methodologically, most multi-objective models optimize profit and emissions without a social metric, while circular economy studies rarely embed variables for perishability or demand volatility. In other words, perishability-focused research argues that a unified performance lens that captures resilience, social justice, and environmental dimension is still missing. Synthesizing the foregoing literature reveals three actionable gaps. First, the interplay between policy incentives and technology adoption remains underexplored; no existing model has integrated green process subsidies and alternative fuel fleets within a single decision space. Second, resilience is poorly understood because empirical evidence outside high-income countries is sparse. Finally, the field lacks a comprehensive objective space that quantifies trade-offs among profit, nutrition, and emissions.

A critical approach to accomplishing these objectives involves the adoption of sustainable and innovative practices within FSCM, as emphasized by Palazzo and Vollero (2022). FSCM plays a pivotal role in coordinating diverse activities across the supply chain, involving a wide range of stakeholders who contribute to and connect through value-adding stages (Bourlakis et al., 2012). These stages include the cultivation, harvesting, and processing of products from agriculture, forestry, and fisheries, as well as their distribution, consumption, and eventual disposal. The decisions made by these stakeholders not only influence one another but also affect broader natural, economic, and

social environments (Nguyen, 2018). The food supply chain network (FSCN), orchestrating the journey of agricultural produce from its origin to the consumer, is characterized by a series of interlinked stages (Shekarabi & Mozdgir, 2025). These networks must integrate dynamic factors such as the perishability of food products, shifting consumer preferences, and environmental considerations including transportation and waste management (Palazzo & Vollero, 2022). Managing these complexities effectively requires a strategic and well-coordinated FSCM, which addresses operational challenges associated with the coordination of product, information, and financial flows among diverse participants (Shekarabi et al., 2024).

A well-structured FSCM framework not only enhances operational efficiency but also plays a key role in promoting food safety, reducing waste, and supporting sustainability goals. The principal areas of focus within FSCM include risk management for perishable goods, innovative technologies that boost supply chain efficiency and transparency, as well as sustainability practices that minimize environmental impact (Cahyadi et al., 2024; Khalid et al., 2024; Panigrahi et al., 2024). FSCM operates as a multi-layered network, managing stages from production and handling to processing, packaging, distribution, and retail. By effectively integrating these stages, FSCM not only ensures a stable and secure food supply but also contributes to establishing a resilient and adaptable system capable of meeting current demands while proactively preparing for future challenges (Bourlakis & Weightman, 2008).

Sustainability in Food Supply Chains

Sustainability within food supply chains has progressed from a specialist topic to a strategic imperative since environmental, social, and economic pressures now shape every stage of production and delivery (Krishnan et al., 2022). Empirical work indicates that lasting improvements only emerge when sustainability principles are woven directly into sourcing, processing, and distribution decisions (Mogale et al., 2023). To illustrate, Munuhwa and Hove-Sibanda (2024) document gains in waste reduction, water stewardship and energy efficiency, explaining how each practice fosters effective waste management and resource conservation. The social dimension is advancing as well, with equity-centered approaches that prioritize nutrient-dense foods for vulnerable communities. Resilience also features prominently: Alabi and Ngwenyama (2023) demonstrate that preparation for unexpected shocks is essential for mitigating disruption-related losses.

Scholars increasingly adopt a triple-bottom-line perspective which balances economic viability, environmental impact, and social equity (Ala et al., 2024). New enablers such as digital technologies, lean methods, and circular

supply chain tools can boost the financial performance, yet they only translate into environmental or social gains when paired with supportive circular processes (Khan et al., 2022).

These findings challenge earlier transaction cost views treating environmental investment solely as an expense driver, highlighting that technology alone seldom guarantees balanced outcomes (Sana, 2022). Integrated models that synchronize sourcing, processing, and routing have demonstrated that greenhouse gas emissions can fall substantially at minimal additional cost when inventory policies are redesigned (Kumar et al., 2025). By contrast, comparable models applied to dairy networks under carbon caps have reported shrinking profit margins though they operate under the same emission constraints (Eskandarpour et al., 2015; Hussein et al., 2021). The divergence stems mainly from system boundary choices. Frameworks that incorporate demand variability and permit flexible transport modes tend to present win–win solutions for cost and emissions (Becerra et al., 2024), while streamlined single-echelon studies often exaggerate the trade-off.

The social pillar remains the least developed. Although researchers apply indicators such as labor safety and nutritional security, there is little agreement on how to weigh or combine them (Desiderio et al., 2022). Optimization models usually proxy social value through employment counts or fairness constraints, an approach that overlooks power imbalances between smallholders and downstream firms and undermines empirical validation. Circular-economy thinking in agri-food settings is also thin, since most studies translate it into operational parameters such as batch size or delivery frequency without fully capturing closed-loop principles (Bigliardi et al., 2024). The absence of integrated modeling masks the interplay among circular strategies, perishability constraints, and inventory-routing policies, a gap that limits both theoretical insight and practical relevance.

Sustainability within food supply chains is conceptualized through three primary dimensions: economic, social, and environmental. Each dimension plays a key role in ensuring supply chains remain equitable, viable, and environmentally friendly (Borsotto et al., 2023; Silva et al., 2023; Yakovleva et al., 2010). However, achieving sustainability is not without its challenges. At every level of the food supply chain, stakeholders confront issues such as limited financial and human resources, shortages of fertilizers and pesticides, as well as insufficient raw materials and other inputs (Lucas et al., 2021; Panigrahy & Panda, 2025). Additional barriers include inadequate information sharing, risks of burglary, and logistical challenges such as marketing difficulties, transportation inefficiencies, and inequitable distribution of consumer prices which are significant for farmers, who experience weakened negotiating

power within the complex echelons of food supply chain networks and struggle with infrastructure constraints, such as limited refrigerated warehousing capacity (Yadav et al., 2022).

Environmental factors further exacerbate these challenges, introducing a wide range of variables that affect food supply chains on local to global scales (Leal Filho et al., 2022; Lucas et al., 2021). Climate-related conditions, such as fluctuations in rainfall, extreme weather events, temperature variations, pest and disease outbreaks, as well as changes in CO₂ and ozone levels, can disrupt both food production and distribution (Davis et al., 2021). These disruptions often result in local shortages of specific foods, which in turn affect food security, especially in regions with limited market access where food production and consumption are closely linked. The perishable nature of food products further complicates supply chain management, as stakeholders have to navigate inconsistent supply and demand, price volatility, and limited information on market conditions (Yadav et al., 2022).

To address these multifaceted challenges, there is a growing interest in adopting circular economy (CE) principles within agri-food supply chains. The circular economy encourages resource circularity, which aligns well with sustainability goals by minimizing waste, maximizing resource efficiency, and enhancing supply chain resilience (Mehmood et al., 2021). As Luo et al. (2022) note, while cost efficiency has historically been a primary performance measure in FSCM, there has been a notable shift toward ascertaining sustainability performance, including strategies to reduce food loss and waste throughout supply chain operations.

Optimization Approaches in Food SCs

Recent research on food supply chains often adopts multi-objective optimization to balance economic costs with environmental impacts such as carbon emissions (Singh & Pandey, 2024). Nevertheless, the relationship between these objectives is far from straightforward. Small changes in transport mode or processing sites can change annual greenhouse gas output considerably even when total financial expenditure stays almost unchanged (Shekarabi et al., 2025). This observation highlights the value of integrated decision frameworks which treat economic and environmental criteria together rather than in isolation (Rohmer et al., 2019). Outcomes also heavily depend on how models handle perishability. When product deterioration is built into the formulation, analysts often uncover solutions that lower both costs and emissions. Excluding perishability, by contrast, tends to expose sharper trade-offs between the two objectives (Wang et al., 2020).



Within this context, multi-objective optimization provides a useful toolkit for navigating the conflicting aims of economic performance, environmental stewardship, and social welfare. Although originally developed for broader applications, these methods transfer well to food systems, helping decision-makers explore the full spectrum of feasible solutions and understand where compromises or synergies arise (Krishnan et al., 2022; Stillitano et al., 2022). Economic viability remains the first and most familiar target, focusing on minimizing costs linked to production, transport, and storage while still fulfilling service and quality requirements.

Hassna et al. (2024) provided a comprehensive analysis of the impact of climate change on global food systems, emphasizing the need to minimize economic costs while maintaining food accessibility under various risk constraints. The second goal of MOO is risk management, dealing with mitigating risks related to climate variability and market volatility. Hassna et al. (2024) and Belamkar et al. (2023) utilized MOO models to demonstrate how they can evaluate and lower risks while boosting supply chain performance under uncertain conditions. The final objective of MOO in food supply chains involves addressing environmental sustainability by managing greenhouse gas emissions and resource depletion. Studies by Cia and Cunha (2024) highlighted the role of logistics in mitigating environmental impacts. Their research designed a sustainable logistics system that revealed how even small reductions in profit margins could result in significant environmental benefits and positive externalities. Further, Jayarathna et al. (2021) analyzed a wide range of studies, concluding that economic and environmental sustainability are critical factors in achieving sustainability in supply chains and logistics.

Various studies have explored the application of MOO in the realm of agricultural contexts. For example, Ackerman Ackerman (2015) developed a green supply chain management framework within the orange juice agro-food cluster, using life cycle assessment and genetic algorithms to minimize environmental impacts while boosting economic profitability. Likewise, Soori et al. (2022) proposed a MOO model for the mushroom industry, indicating that investment in oyster mushrooms could improve social and economic outcomes, whereas investment in button mushrooms could enhance all dimensions of sustainability. Balancing multiple objectives, such as cost minimization, risk mitigation, and maximizing nutritional value, is a critical aspect of MOO. Hassna et al. (2024), in their study, offered a detailed exploration of how climate change affects global food systems and how optimized food supply chains can promote resilience. Another important aspect of MOO is its ability to integrate uncertainty into planning processes. This capability helps decision-makers choose

strategies that respond effectively to market fluctuations, thereby ameliorating supply chain resilience and sustainability (Esteso et al., 2024; Soori et al., 2022). Esteso et al. (2024) argued that inadequate planning, such as planting and harvesting without considering broader supply chain factors, can lead to imbalances between supply and demand and lead to price volatility. They developed a MOO model for managing agricultural processes from planting and harvesting to storage, packaging, and sales.

Case studies and practical applications further manifest the value of MOO in sustainable food supply chains. Ackerman's (2015) research on the orange juice supply chain proposed integrating eco-labeling into supply chains, which would enhance sustainability. Soori et al. (2022) also highlighted the practical benefits of MOO in the mushroom industry, where the model contributed to enhanced sustainability outcomes. A sustainable food supply chain typically involves three interconnected levels: i) the primary production level (farmers), ii) the processing level (manufacturers), and iii) the distribution level (wholesalers and retailers). Integrating MOO techniques with this three-level model helps balance the goals of economic, social, and environmental sustainability, ensuring that supply chains are resilient, adaptable, and capable of addressing global challenges such as climate change and food insecurity. The role of sustainable food supply chains extends to minimizing environmental impacts while promoting economic viability and social welfare. Fulfilling these goals requires collaboration across all supply chain levels. Research by Salisu and Žilinskienė (2021) as well as Palazzo and Vollero (2022) emphasized the need for embedding sustainability into each stage of the supply chain to build resilience against global disruptions. Studies by Latino et al. (2021) along with Salisu and Žilinskienė (2021) demonstrated that short food supply chains could improve economic sustainability by lowering transportation costs and increasing market access for farmers.

Flexible Systems Management in Supply Chains

Flexible systems management regards a supply network as an open socio-technical system that should sense change, learn, and adapt with minimal delay. Within this perspective, SC flexibility is defined as the capacity of a firm to adjust, adapt, and transform its resources as well as processes together with partners so that the network can cope with external impacts (Varma et al., 2024). SC flexibility can be defined as a response to rising uncertainty and identified demand variation, supplier unreliability, and competitive actions as principal causes that prompt managers toward flexible design selections (Song, et al., 2018). This concept can also be unpacked into complementary

dimensions that operate at strategic, tactical, and operational levels (Agrawal et al., 2023).

Research has highlighted flexibility in capacity, production processes, supplier collaboration, and delivery operations as the principal drivers of performance improvement (Enrique et al., 2022). When organizations can adjust volume and work with suppliers, they achieve superior cost efficiency and service levels compared to less adaptable peers. Further, applying multiple flexibility dimensions across the network enhances responsiveness, reduces inventory risk, and boosts customer satisfaction (Jafari et al., 2023). There is consensus on knowledge management and digital technologies as critical enablers of flexible systems. Centralized information platforms, collaborative decision spaces, and supplier relationship portals reduce decision latency and integrate dispersed data, improving the network's ability to reconfigure quickly (Haleem et al., 2018).

Big-data analytics, additive manufacturing, and cyber-physical production lines further loosen the traditional link between volume and cost, enabling planners to switch products, routes, or capacities without incurring prohibitive penalties (Srivastava & Bag, 2023). Along the COVID-19 crisis, firms combining real-time information exchange with process-switching options restored operations faster and protected revenue streams more effectively than rigid competitors. One of the consistent themes across the literature is the role of flexibility in risk mitigation and resilience building (Rani & Joshi, 2024; Settembre-Blundo et al., 2021).

Analysis of how built-in options shape resilience indicates that having multiple suppliers, dual sourcing agreements, and strategic postponement points create buffers that absorb fluctuations and major shocks alike (Singh et al., 2021). Evidence from real-world disruptions suggests that networks designed with these options bounce back faster and maintain stronger financial performance even when external conditions such as currency swings become volatile (Settembre-Blundo et al., 2021). Flexibility also proves essential for meeting environmental goals, since organizations can introduce low-carbon or circular processes without slowing their response to market changes. Over time the field has moved beyond simple definitions and classifications to embrace richer, system-level frameworks (Paul & Saha, 2025; Varma et al., 2024). Early work introduced tools such as balanced scorecard measures to capture flexibility in concrete terms. More recent analyses highlight the influence of digital innovations, particularly Industry 4.0 technologies, on adaptive capacity and call for studies that tie front-line capabilities to network-wide results (Höse et al., 2023).

Sustainability in Flexible Systems Management

The rapid acceleration of market volatility and environmental imperatives has elevated sustainability to a core design principle within flexible systems management (Singh et al., 2021). Supply chains must now arrange capabilities that redesign resources and processes not only to absorb disruptions but also to advance carbon reduction and social equity goals (Varma et al., 2024). Recent studies highlight how digital technologies such as blockchain traceability, AI-driven demand sensing, and IoT-enabled visibility create a feedback loop that strengthens both agility and ecological performance (Srivastava et al., 2025). By unifying resilience and green objectives, firms can pivot in real time to optimize routing for lower emissions or reconfigure production to favor renewable-powered facilities (Gholizadeh et al., 2025).

Key enablers of this integrated paradigm include stakeholder collaboration, transparency, and risk-bearing capacity. An analysis identified fourteen critical factors, from regulatory alignment and financial stability to supply chain risk management, that collectively underpin the sustainable resilience of flexible networks (Mangla et al., 2014). Complementary work on green supply chain flexibility distilled a set of enablers and barriers such as cost premium trade-offs, data-sharing hurdles, and governance commitment, underscoring the need for cross-functional coordination to sustain triple-bottom-line targets (Varma et al., 2024).

Advancing this field requires elevating sustainability metrics to first-class objectives and embedding them in real-time decision platforms (Daraio et al., 2025). A novel research direction proposes coupling multi-objective metaheuristics with digital twin environments, enabling the exploration of near-Pareto trade-offs among cost, carbon intensity and social welfare under stochastic scenarios. Such a holistic, data-driven framework equips managers to calibrate green incentives and flex capacity responsively, forging supply networks that are both agile and ecologically restorative.

Research Gap

A comprehensive analysis of the literature indicates three critical research gaps that delineate a clear agenda for advancing sustainable food supply chain studies.

1. Policy incentives and technology adoption

Prior studies have examined either producer subsidies for green processes or financial support for alternative fuel vehicles in isolation. There is no integrated model that considers both levers inside one decision space.

2. Multi-echelon and perishability



Most optimization research continues to focus on one- or two-tier supply chains, often modeling products as non-perishable flows. In contrast, the managerial reality of food supply networks involves at least three interconnected echelons, all being subject to strict shelf-life constraints. The lack of structural and temporal detail in existing models limits the external validity of their findings and leaves critical issues—such as inventory aging, consolidation economies, and cross-tier spillover effects—largely unaddressed.

3. Partial coverage of the triple-bottom-line objective space

While most existing models prioritize economic and environmental objectives, social equity is typically represented through coarse proxies such as job creation. This study demonstrates that subsidy thresholds can boost profits but may also exacerbate emissions or equity disparities. Nevertheless, current theoretical frameworks lack a unified set of metrics capable of rigorously quantifying these trade-offs.

Proposed Multi-Objective Model

This study presents a multi-objective mathematical model for a sustainable three-level food supply chain, including producers, distributors, and retailers. Three sustainability dimensions were considered: economic, environmental, and social. The economic dimension covers production costs, inventory maintenance, distribution costs, overhead, etc., while the environmental dimension captures the carbon emissions in the supply chain from production and distribution activities. The social dimension is represented by the proportion of products manufactured using green processes, reflecting public health benefits and social welfare improvements associated with green manufacturing methods.

The case study is a dairy products company. To solve the sustainable supply chain design problem, we employ both exact and metaheuristic methods: the model is implemented as a mixed-integer program solved by the commercial solver, and a Non-dominated Sorting Genetic Algorithm II (NSGA-II) is applied to efficiently obtain near-optimal solutions. These solution methods account for government subsidy policies, including production subsidies for green products and per-use subsidies for AFVs. Government incentives are modeled via parameters that lower the effective costs, such as a subsidy per green product and a per-vehicle-use subsidy for green transport.

Designing supply chains that balance economic profitability with social and environmental objectives has received considerable attention in recent years. Many previous studies have considered simpler two-level models. However, food supply chains often include an intermediate

distributor (wholesale) stage. Introducing a distributor allows consolidation of shipments, improved delivery efficiency, and more realistic modeling of the distribution network. Thus, our model utilizes a three-level structure, as displayed in Fig. 1. This structure enables economies of scale in logistics, potentially lowering total costs and emissions.

Model Assumptions and Notations

The designed model has some assumptions as follows:

- Parameters such as travel costs between members at different levels of the supply chain, demands, and carbon emission rates for each member and vehicle are given and fixed.
- The cargo transportation capacity of each type of vehicle is given and fixed.
- The potential locations for distribution centers are fixed and predetermined.
- The pollution caused by green production processes is far less than that from conventional production.
- We assume a core producer at the upper level who plays a centralized supply chain management role.
- Unfulfilled demand can be considered as delayed orders to the retailer with associated costs.
- The proposed model is three-level, multi-period, and multi-product.
- Regardless of the production process, the producer offers products at a single price and through a single distribution channel.
- The government provides per-unit production subsidies for green products and per-use subsidies for green vehicles, incorporated via the model's subsidy parameters.

Sets

$i \in N$	Set of distributors, indexed by i .
$p \in P$	Set of products, indexed by p .
$v \in V$	Set of vehicles, indexed by v .
$r \in R$	Set of retailers, indexed by r .
$k \in K$	Set of types of production processes, indexed by k .
$t \in T$	Set of time units, indexed by t .

Parameters

UC_p	Cost of producing one unit of product p .
TC_v	Transportation cost per unit by vehicle v .
A_i	Distance to distributor i .
A_{ir}	Distance from distributor i to retailer r .
FV_v	Fixed cost of using vehicle v .
FI_i	Fixed cost of setting up a distribution center at distributor i .
$CapV_v$	Transportation capacity of vehicle v .

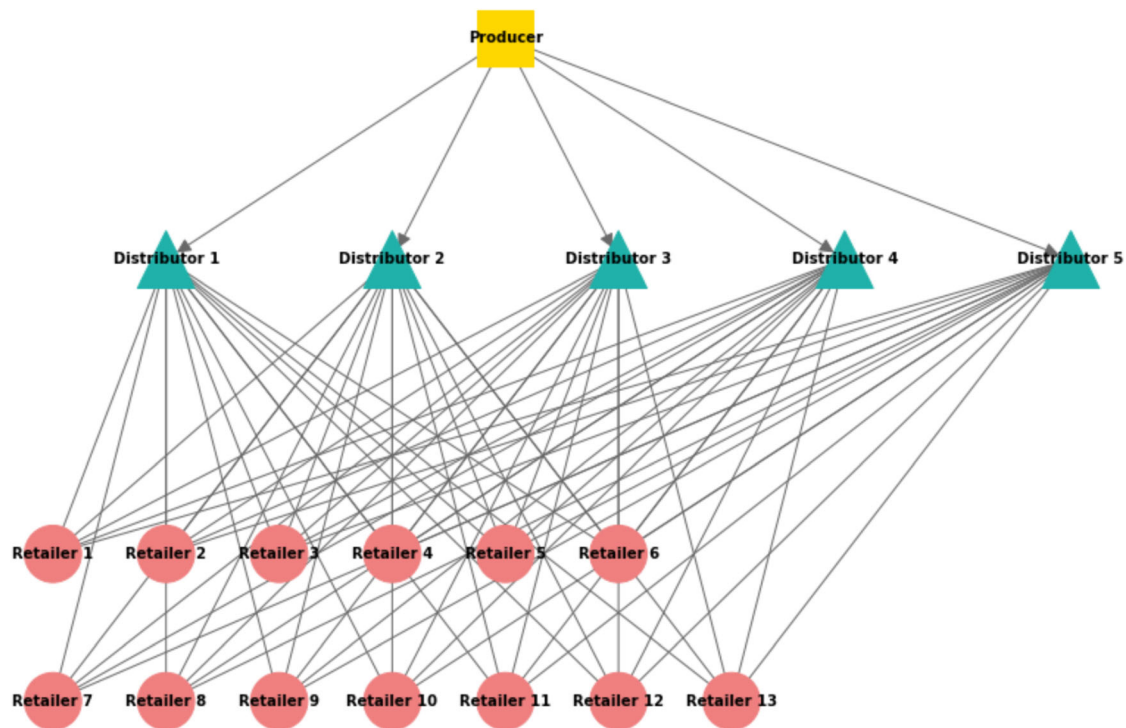


Fig. 1 Conceptual diagram of the three-level supply chain

d_{prt}	Demand of retailer r for product p in period t .
<i>Sets</i>	
$i \in N$	Set of distributors, indexed by i .
$p \in P$	Set of products, indexed by p .
$v \in V$	Set of vehicles, indexed by v .
$r \in R$	Set of retailers, indexed by r .
$k \in K$	Set of types of production processes, indexed by k .
$t \in T$	Set of time units, indexed by t .

Parameters

UC_p	Cost of producing one unit of product p .
TC_v	Transportation cost per unit by vehicle v .
A_i	Distance to distributor i .
At_{ir}	Distance from distributor i to retailer r .
FV_v	Fixed cost of using vehicle v .
FI_i	Fixed cost of setting up a distribution center at distributor i .
$CapV_v$	Transportation capacity of vehicle v .
d_{prt}	Demand of retailer r for product p in period t .
BC_{prt}	Cost of delayed order for retailer r for product p in period t .
ED_{pv}	The amount of carbon emitted per unit distance and weight by vehicle v for product p .
EP_{pk}	The amount of carbon emitted per unit of product production for product p using production process k .
$CapM$	

Production capacity of a variety of products by the producer in one work shift.

θ_{pk} Excess cost to produce one unit of product p using production process k .

λ_{pk} Subsidy (government incentive) received for producing one unit of product p using production process k .

γ_v Subsidy (government incentive) for using vehicle v .

$IM0_p$ Initial inventory of product p in the producer's warehouse.

$ID0_{pi}$ Initial inventory of product p in the distributor warehouse i .

$IR0_{pr}$ Initial inventory of product p in the retailer warehouse r .

H_p Maintenance cost per unit of product p in the producer's warehouse.

H_{pi} Maintenance cost per unit of product p in the warehouse of distributor i .

H''_{pr} Maintenance cost per unit of product p in a retailer warehouse r .

Variables

x_{pkt} Quantity of product p produced using process k in period t .

y_{pivt} Quantity of product p transported by vehicle v to distributor i in time unit t .

z_{pirt}

	Quantity of product p transported by vehicle v from distributor i to retailer r in time unit t .
IM_{pt}	Inventory of product p in the producer's warehouse at the end of period t .
ID_{pit}	Inventory of product p in distributor i 's warehouse at the end of period t .
IR_{prt}	Inventory of product p in retailer r 's warehouse at the end of period t .
f_v	Binary variable that takes the value of 1 if vehicle v is used, and 0 otherwise.
e_i	Binary variable that takes the value of 1 if distributor i is set up, and 0 otherwise.
β_{prt}	The amount of delayed order of product p for retailer r over period t .
BC_{prt}	Cost of delayed order for retailer r for product p in period t .
ED_{pv}	The amount of carbon emitted per unit distance and weight by vehicle v for product p .
EP_{pk}	The amount of carbon emitted per unit of product production for product p using production process k .
$CapM$	Production capacity of a variety of products by the producer in one work shift.
θ_{pk}	Excess cost to produce one unit of product p using production process k .
λ_{pk}	Subsidy (government incentive) received for producing one unit of product p using production process k .
γ_v	Subsidy (government incentive) for using vehicle v .
$IM0_p$	Initial inventory of product p in the producer's warehouse.
$ID0_{pi}$	Initial inventory of product p in the distributor warehouse i .
$IR0_{pr}$	Initial inventory of product p in the retailer warehouse r .
H_p	Maintenance cost per unit of product p in the producer's warehouse.
HI_{pi}	Maintenance cost per unit of product p in the warehouse of distributor i .
H''_{pr}	Maintenance cost per unit of product p in a retailer warehouse r .

Variables

x_{pkt}	Quantity of product p produced using process k in period t .
y_{pivt}	Quantity of product p transported by vehicle v to distributor i in time unit t .
z_{pirvt}	Quantity of product p transported by vehicle v from distributor i to retailer r in time unit t .
IM_{pt}	Inventory of product p in the producer's warehouse at the end of period t .

ID_{pit}	Inventory of product p in distributor i 's warehouse at the end of period t .
IR_{prt}	Inventory of product p in retailer r 's warehouse at the end of period t .
f_v	Binary variable that takes the value of 1 if vehicle v is used, and 0 otherwise.
e_i	Binary variable that takes the value of 1 if distributor i is set up, and 0 otherwise.
β_{prt}	The amount of delayed order of product p for retailer r over period t .

Mathematical Model

A three-objective mathematical model is presented for the problem. The first objective function minimizes the total cost of the supply chain, the second objective function minimizes the emission of pollutants in the production as well as distribution sectors of the supply chain, and the third objective function minimizes the ratio of products produced through the green process to total products produced, such that it considers operations constraints of the problem. As the problem considered in this study has three objective functions, a MOO approach is utilized and a multi-objective mathematical model is presented for the problem in the form of a mixed-integer nonlinear programming model (MINLP).

$$\begin{aligned} \text{Min } Obj_1 = & \sum_{p \in P} \sum_{k \in K} \sum_{t \in T} (UC_p + \theta_{pk} - \lambda_{pk}) \cdot x_{pkt} \\ & + \sum_{p \in P} \sum_{i \in N} \sum_{v \in V} \sum_{t \in T} TC_{pv} \cdot A_i \cdot y_{pivt} \\ & + \sum_{p \in P} \sum_{i \in N} \sum_{v \in V} \sum_{r \in R} \sum_{t \in T} TC_{pv} \cdot G_{ir} \cdot z_{pirvt} \\ & + \sum_{p \in P} \sum_{r \in R} \sum_{t \in T} BC_{prt} \cdot \beta_{prt} + \sum_{p \in P} \sum_{t \in T} H_p \cdot IM_{pt} \\ & + \sum_{p \in P} \sum_{i \in N} \sum_{t \in T} H_{pr} \cdot ID_{pit} \\ & + \sum_{p \in P} \sum_{r \in R} \sum_{t \in T} H''_{pr} \cdot IR_{prt} + \sum_{i \in N} FI_i \cdot e_i \\ & + \sum_{v \in V} (FV_v - \gamma_v) \cdot f_v \end{aligned} \quad (1)$$

$$\begin{aligned} \text{Min } Obj_2 = & \sum_{p \in P} \sum_{k \in K} \sum_{t \in T} EP_{pk} \cdot x_{pkt} \\ & + \sum_{p \in P} \sum_{i \in N} \sum_{v \in V} \sum_{t \in T} ED_{pv} \cdot A_i \cdot y_{pivt} \\ & + \sum_{p \in P} \sum_{i \in N} \sum_{v \in V} \sum_{r \in R} \sum_{t \in T} ED_{pv} \cdot G_{ir} \cdot z_{pirvt} \end{aligned} \quad (2)$$

$$\text{Max } Obj_3 = \sum_{p \in P} \sum_{k \in K} \sum_{t \in T} \frac{x_{p1t}}{x_{pkt}} \quad (3)$$

S.t:

$$\sum_{p \in P} \sum_{k \in K} x_{pkt} \leq CapM \quad \forall t \in T \quad (4)$$

$$IM_{p1} = IM0_p + \sum_{k \in K} x_{pk1} - \sum_{i \in N} \sum_{v \in V} y_{piv1} \quad \forall p \in P \quad (5)$$

$$IM_{pt} = IM_{pt-1} + \sum_{k \in K} x_{pkt} - \sum_{i \in N} \sum_{v \in V} y_{pivt} \quad \forall p \in P, \quad t \in T \quad (6)$$

$$\sum_{p \in P} y_{pivt} \leq CapV_v \cdot f_v \quad \forall v \in V, \quad i \in N, \quad t \in T \quad (7)$$

$$ID_{pi1} = ID0_{pi} + \sum_{v \in V} y_{piv1} - \sum_{r \in R} \sum_{v \in V} z_{pirv1} \quad \forall p \in P, \quad i \in N \quad (8)$$

$$ID_{pit} = ID_{pit-1} + \sum_{v \in V} y_{pivt} - \sum_{r \in R} \sum_{v \in V} z_{pirvt} \quad \forall p \in P, \quad i \in N, \quad t = \{2, 3, \dots, |T|\} \quad (9)$$

$$\sum_{p \in P} z_{pirvt} \leq CapV_v \cdot f_v \quad \forall i \in N, \quad r \in R, \quad v \in V, \quad t \in T \quad (10)$$

$$IR_{pr1} = IR0_{pr} + \sum_{i \in N} \sum_{v \in V} z_{pirv1} - d_{pr1} \quad \forall p \in P, \quad r \in R \quad (11)$$

$$IR_{prt} + \beta_{prt} = IR_{prt-1} + \sum_{i \in N} \sum_{v \in V} z_{pirvt} - \beta_{prt-1} - d_{prt} \quad \forall p \in P, \quad i \in N, \quad t = \{2, 3, \dots, |T|\} \quad (12)$$

$$\sum_{p \in P} \sum_{v \in V} \sum_{t \in T} y_{pivt} \leq M \cdot e_i \quad \forall i \in N \quad (13)$$

$$\sum_{p \in P} \sum_{r \in R} \sum_{v \in V} \sum_{t \in T} z_{pirvt} \leq M \cdot e_i \quad \forall i \in N \quad (14)$$

$$x_{pkt}, y_{pivt}, z_{pirvt}, IM_{pt}, ID_{pit}, IR_{prt}, \beta_{prt} \geq 0 \quad e_i, f_v = \{0, 1\} \quad (15)$$

Equation (1) as the first objective function of the mathematical model seeks to minimize the total cost in the supply chain including the cost of transportation from producer to distributors, cost of transportation from distributor to retailers, cost of a delayed order, cost of inventory storage in producer's warehouse, the cost of maintaining inventory in the distributors' warehouse, the cost of maintaining inventory in the retailer's warehouse, the fixed cost of initiating a distributor, and the fixed cost of using a vehicle. Equation (2) as the second objective function of the mathematical model is one of the sustainability criteria of the problem intending to minimize the emission of environmental pollution related to production, upstream transportation fleet (producer to distributors), and downstream transportation fleet

(distributors to retailers). Equation (3) as the third objective function of the mathematical model is another sustainability criterion of the supply chain as a measure for the social level of the supply chain and maximizes the ratio of products produced by the green method to total products produced.

Constraint (4) states that the total amount of production per work shift cannot exceed the producer's capacity in one shift. Constraint (5) determines the initial inventory of the producer's warehouse. Constraint (6) calculates the inventory of every product in each period at the producer's warehouse. Constraint (7) ensures that the goods transported to each distributor by vehicle v in every period cannot exceed that vehicle transportation capacity. However, products can be transported from a producer to the distributors by a vehicle when that vehicle is employed. Constraint (8) determines the initial inventory of each product in each distributor's warehouse. Constraint (9) calculates product inventory p at the warehouse of each distributor i in each period t . Constraint (10) describes that the number of goods transported from a distributor to a retailer per period by each vehicle v should be less than or equal to the capacity of that vehicle. Further, products can be transported from the distributor to the retailers by vehicle when that vehicle is utilized. Constraint (11) indicates the initial inventory of each product p in each period at the warehouse of each retailer r . Constraint (12) calculates the inventory of the product p at the retailer r in each period t . Constraint (13) states that goods can be sent to a distributor only when that distributor is set up. Constraint (14) states that goods can be sent from the distributor i to retailers when that distributor is set up. Constraint (15) specifies the appropriate and allowed values for the variables.

Linearization

The proposed model is a MINLP given the nonlinear relationship in the third objective function. Since solving nonlinear problems adds to the difficulty of the problem, in this section, we replace the relation of the third objective function employed in the model with an equivalent linear relationship. The third objective function maximizes the ratio of production of the products produced using the green process to the total number of products produced.

As the products produced by the green process are less harmful to public health than conventional products, the products produced more by the green process claim a larger share in total production. On the other hand, given the hypothesis considered, irrespective of the type of production process (green or normal) to maintain competitive advantage, a product has the same price and a single distribution channel. Thus, customer demand can be covered



by both types of green or normal products. In other words, there is a demand for $p \in P$ products regardless of the type of $k \in K$ process utilized in its production. For this reason, augmenting the share of the product produced with the green process can enhance the general health of people in the community, which is in line with social concerns and a subset of criteria linked to a sustainable supply chain. Hence, according to the above-mentioned explanations and considering the point that both types of green and ordinary products play a role in meeting demand, maximizing the number of goods produced by the green method and process will lead to an increasing share of green products in general. Thus, the nonlinear relationship in the third objective function is replaced by the relation (16).

$$\text{MaxObj}_3 = \sum_{p \in P} \sum_{t \in T} x_{p1t} \quad (16)$$

Relation (16) maximizes the production of products using the green production process, which is a linear relationship and it was shown to be equivalent to Eq. (3), in such a way that the maximization of Relation (16), that is, the produced products using the green process, can lead to the maximization of the relations (3). The mixed-integer linear model presented in this section includes Relations (11)–(2) as well as Constraints (4)–(15) as well as the objective function of Relation (16).

While this linear approximation transforms the original nonlinear ratio into a tractable mixed-integer model, it also assumes that each additional unit of green production contributes the same social benefit. In real-world situations, community health gains and consumer well-being often present diminishing returns once a certain share of green products is reached. Capturing this effect would require nonlinear or piecewise social-benefit functions, which in turn introduce additional binary variables or nonlinear constraints and markedly increase computation time. We therefore adopt the linear form as a pragmatic choice to preserve problem scale and solution speed, while still indicating the main policy and operational interactions.

Solution Approach

Experimental Example

We now classify example sizes into small (Examples 1–10), moderate (Examples 11–20), and large (Examples 21–30). A small example has up to 2 products and ≤ 17 retailers; a moderate example includes 3–6 products and 18–50 retailers; a large example involves ≥ 7 products and > 50 retailers, with these scales reported in Table 1. All calculations were carried out on a Core i7-4510U laptop with a 2.00–2.26 GHz CPU.

Since the proposed model is MILP, CPLEX was chosen to code and solve the problem. On the other hand, since the model is multi-objective, the ϵ -constraint method is applied. Table 2 outlines the results related to the number of green and non-green products produced in solving experimental examples. Empty entries in Table 2 mean that the corresponding product has not been modeled in that example. We evaluate the model under three objective functions, labeled Objectives 1–3 in Table 3. Objective 1 minimizes the total cost, Objective 2 minimizes carbon emissions, and Objective 3 maximizes the social/health index (i.e., the green product ratio). These objectives capture different policy priorities and will be discussed in the context of the results.

According to Table 2, since the surplus cost paid for the production of the product using the green production process is greater than the subsidy received for it, it is not a significant number against the unit cost of production. Thus, for all experimental examples, the model has decided to meet a greater amount of demand with non-green (normal) products. Hence, the marginal cost versus subsidy gap suppresses green product uptake. Increasing the per-unit subsidy, or reducing the excess cost, shifts the optimum toward a higher share of green products, a policy lever discussed in section “Sensitivity Analysis on θ : Insights and Managerial Implications”. Also, since the third objective function is to maximize the total product produced using the green process, the sum of the series of products produced using the green production process is equal to the value of the third objective function for each of the experimental examples.

Table 3 presents a clear trend regarding the scalability of both the distribution network design and solver performance as the size of the experimental examples grows. Solver runtime rises nonlinearly, from under 5 s for the smallest examples to nearly 3000 s for the largest, reflecting the exponential increment in combinatorial complexity when more products, retailers, and decision variables are introduced. Despite this growth, commercial solver maintains robust convergence; small examples solve to optimality (0% gap) in seconds, while moderate examples reach residual gaps below 0.25% in minutes, and large examples stabilize around a 1% gap within the two-to-three-hour window. This performance profile indicates that the mixed-integer structure, dominated by binary DC-opening and vehicle selection decisions combined with continuous flow variables, remains tractable for real-world scales when applied with a modern solver and the ϵ -constraint approach.

In parallel, the number of distribution centers selected by the model increases modestly with example size, from a single DC in the simplest example to up to nine in the most complex, highlighting the role of fixed-opening costs as a

gating factor. The fact that five candidate DCs are never opened across all 30 examples stems from a critical threshold; only those centers whose demand volumes compensate their fixed costs, such as where the fixed cost per unit served falls below the average transport cost saving per unit, remain operational. This threshold-driven mechanism ensures that infrastructure expansion is controlled, mirroring strategic capital deployment decisions in large-scale logistics networks and preventing overinvestment in underutilized facilities.

Considering vehicle utilization, the results reveal a systematic shift from predominantly fossil vehicles in smaller examples to hybrid and alternative fuel vehicles as government subsidies and green mandates intensify. However, even in the largest examples where multiple AFVs are feasible, the model only deploys them when their marginal operating cost (net of subsidy) undercuts that of

conventional fleets, reflecting a marginal cost versus subsidy equilibrium which governs green technology adoption. This nuanced, cost-driven allocation suggests the necessity of sufficiently large subsidies to tip the balance in favor of greener options.

Comparison Test

The CPLEX solver was utilized to tackle problems of different scales. Further, the NSGA-II, a multi-objective genetic metaheuristic algorithm, was employed for experimental cases. For NSGA-II, we performed tuning and selected a population size of 250, crossover probability of 0.8, mutation probability of 0.2, and 500 generations to balance convergence quality against computation time. Subsequently, a comparative analysis was undertaken to evaluate the outcomes derived from both the CPLEX solver and the Adapted NSGA-II algorithm. Table 4 summarizes the findings obtained from solving all experimental examples, and Table 5 outlines a detailed computation time comparison for the large-scale examples for Examples 21–30.

The results of Table 4 reveal a correlation between the size of the experimental examples and the complexity of the supply chain network problems. Specifically, as the number of retailers and the variety of products in the supply chain network increase, so does the complexity of the problem accordingly. This augmentation of complexity means that employing exact methodologies to solve larger-scale problems might require significantly extended time frames, potentially spanning days, weeks, or even months.

The results in Table 5 present an initially high computational advantage of the adapted NSGA-II over CPLEX for large-scale problems, for Examples 21–23, runtime reductions exceed 82%. As the problem dimensions expand further, Examples 24–30, the observed speed-up gradually declines to around 70–77%. This tapering effect reflects the fixed-size population and generation count imposing a search budget ceiling; once the Pareto front's complexity surpasses a certain point, additional problem scale leads to diminishing returns in runtime reduction.

In spite of this slowdown in relative speed-up, solution quality remains robust; the quality metric (QM) never falls below 37.8% and averages above 60% even in the hardest examples, suggesting that the heuristic reliably approximates the true Pareto front. The convergence metric (CM) stays under 1.05%, representing graceful convergence degradation, while the diversity metric (DM) oscillates in response to front curvature changes, dipping below 1% when the solution set becomes narrow, such as Example 24, and rising above 3% when the front shape is more irregular; it can be observed in Examples 21–23.

Table 1 Size of experimental examples

Example	Number of products produced	Number of retailers
1	1	5
2	2	8
3	2	10
4	2	12
5	2	14
6	2	17
7	2	22
8	2	23
9	3	25
10	3	27
11	3	28
12	4	30
13	4	32
14	4	34
15	4	36
16	5	38
17	5	42
18	5	44
19	6	46
20	6	48
21	6	50
22	7	52
23	7	55
24	7	58
25	7	60
26	8	62
27	8	64
28	8	66
29	8	68
30	8	70



Table 2 Results related to the amount of product produced in each experimental example

Example	Number of green products produced								Number of non-green products produced							
	P1	P2	P3	P4	P5	P6	P7	P8	P1	P2	P3	P4	P5	P6	P7	P8
1	–	–	–	–	–	–	–	–	25,410	–	–	–	–	–	–	–
2	–	–	–	–	–	–	–	–	43,232	45,111	–	–	–	–	–	–
3	55,511	–	–	–	–	–	–	–	–	55,816	–	–	–	–	–	–
4	67,231	–	–	–	–	–	–	–	–	65,128	–	–	–	–	–	–
5	–	78,103	–	–	–	–	–	–	78,019	–	–	–	–	–	–	–
6	–	–	–	–	–	–	–	–	99,497	96,242	–	–	–	–	–	–
7	–	129,426	–	–	–	–	–	–	127,080	–	–	–	–	–	–	–
8	–	137,539	–	–	–	–	–	–	132,405	–	–	–	–	–	–	–
9	–	–	–	–	–	–	–	–	145,785	144,876	143,899	–	–	–	–	–
10	–	–	–	–	–	–	–	–	156,302	157,177	158,268	–	–	–	–	–
11	163,504	–	–	–	–	–	–	–	–	163,199	164,160	–	–	–	–	–
12	179,482	–	172,984	–	–	–	–	–	–	172,489	–	172,249	–	–	–	–
13	–	–	–	–	–	–	–	–	189,282	185,638	185,933	183,291	–	–	–	–
14	–	199,994	–	–	–	–	–	–	202,142	196,671	196,846	–	–	–	–	–
15	–	–	–	–	–	–	–	–	214,156	212,285	150,679	209,419	–	–	–	–
16	–	–	214,206	–	–	–	–	–	217,802	224,209	–	215,551	223,526	–	–	–
17	248,080	256,761	244,098	–	71,419	–	–	–	–	–	–	243,442	184,261	–	–	–
18	265,544	–	–	–	–	–	–	–	–	260,771	268,601	255,616	255,655	–	–	–
19	273,802	–	273,584	–	–	272,502	–	–	–	267,137	–	270,951	274,083	–	–	–
20	284,861	283,586	–	–	284,050	–	–	–	–	–	283,866	277,197	–	283,635	–	–
21	–	295,428	–	–	295,509	289,847	–	–	295,471	–	297,867	297,953	–	–	–	–
22	–	–	–	–	302,798	–	315,818	–	313,450	297,718	305,181	309,287	–	309,958	–	–
23	–	–	319,134	–	–	–	–	–	334,708	332,138	–	324,909	325,184	320,486	326,245	–
24	–	–	–	–	–	343,581	337,717	–	344,272	352,560	351,598	336,628	345,442	–	–	–
25	–	–	–	–	362,488	–	–	–	357,598	325,234	349,303	359,204	–	363,590	348,444	–
26	352,859	–	–	–	–	–	–	–	–	365,303	370,879	368,731	363,084	372,855	379,571	369,813
27	–	–	–	–	364,652	–	–	–	375,505	385,075	391,007	384,559	–	376,188	382,224	379,054
28	–	–	–	–	389,817	–	397,463	–	375,487	386,714	393,566	390,792	–	394,397	402,215	–
29	401,390	–	412,992	–	–	–	–	–	–	405,006	–	411,073	388,481	399,288	405,698	402,928
30	–	410,910	418,126	415,537	–	–	425,503	420,070	401,021	–	–	–	412,147	415,537	–	–

Table 3 Results of solving experimental examples

Example	Objective function 1	Objective function 2	Objective function 3	Number of distributors	Vehicles			Time (s)	Gap %
					Fossil	Hybrid	Alternatives		
1	9,157,180.65	2,217,880.84	0	1	1	–	–	3.2	0
2	17,299,923.15	6,136,220.33	0	3	–	1	–	4.7	0
3	22,462,120.60	11,004,563.43	55,511	5	–	–	1	6.5	0
4	28,647,839.65	6,832,694.46	67,231	5	–	1	–	8.4	0
5	33,813,132.42	10,372,890.99	78,103	5	–	1	–	9.3	0
6	40,352,266.62	15,178,115.80	0	5	–	1	1	11.7	0
7	51,577,525.79	22,943,590.80	129,426	5	–	1	1	13.9	0.01
8	58,170,295.12	17,462,012.69	137,538	5	–	1	1	15.2	0.01
9	153,446,310.00	45,786,177.33	0	5	1	1	1	120.5	0.05
10	161,771,356.35	43,576,875.53	0	6	–	1	1	142.3	0.09
11	173,081,731.12	37,145,730.14	163,504	6	1	1	1	178.6	0.14
12	291,815,928.65	46,155,968.39	352,466	7	1	–	–	225.4	0.16
13	307,806,621.06	52,878,971.54	0	7	1	–	1	258.9	0.12
14	339,698,563.52	41,807,624.30	199,994	7	1	–	–	300.7	0.19
15	358,411,720.12	49,062,474.73	0	7	1	1	–	345.2	0.22
16	552,431,019.62	51,114,154.68	214,206	7	1	1	1	390.6	0.18
17	649,286,178.27	104,137,988.72	820,359	6	1	1	1	835.4	0.27
18	681,428,008.68	106,858,329.89	265,544	7	1	1	1	912.1	0.31
19	712,360,368.66	133,704,133.38	547,387	7	1	1	1	1,040.8	0.55
20	746,356,146.69	105,364,497.36	852,498	5	1	1	1	1,225.6	0.58
21	766,382,370.13	80,135,574.24	590,938	7	1	1	1	1,358.3	0.66
22	821,069,296.09	119,344,188.66	302,798	6	1	1	1	1,540.9	0.73
23	871,024,475.73	209,309,469.49	319,134	7	1	1	1	1,672.4	0.81
24	961,058,210.35	130,761,136.20	681,299	7	1	1	1	1,810.5	0.90
25	996,219,182.05	165,173,668.36	362,488	8	1	1	1	1,975.3	0.98
26	1,318,700,169.43	237,791,844.54	352,859	8	1	1	1	2,150.8	1.05
27	1,381,588,350.28	218,044,419.45	364,651	6	1	1	1	2,324.5	1.02
28	1,409,245,660.17	205,925,221.91	787,280	7	1	1	1	2,499.1	1.07
29	1,461,802,925.65	213,870,108.54	814,382	9	1	1	1	2,668.7	1.07
30	1,476,326,652.41	310,195,735.75	2,090,147	6	1	1	1	2,832.6	1.06

These findings demonstrate that NSGA-II's fixed-parameter design delivers a pragmatic balance between computation time and solution optimality for real-world supply chain network design.

Case Study, Discussion, and Sensitivity Analysis

The case study centers on a dairy products company operating within a supply chain network. This network comprises 70 retailers and 20 suppliers, handling 8 distinct dairy products over a planning horizon of 10 time periods. The model integrates various operational and environmental parameters, including production costs,

transportation costs, and emission levels. It incorporates government subsidies with the aim of promoting green production processes and the use of alternative fuel vehicles. In this way, it evaluates the dual impact of these subsidies on overall cost efficiency and emission reduction. This study not only provides an understanding of the economic benefits associated with sustainable practices but also demonstrates how strategic policy interventions can facilitate the transition toward a greener supply chain.



Sensitivity Analysis on λ_{pk} : Insights and Managerial Implications

The sensitivity analysis of the λ_{pk} , a key determinant in supply chain management which models the subsidy level for green production processes, as outlined in Table 6, represents the subsidies for producing products using different production processes. We explored its influence on total costs, emissions, and green production by varying the λ_{pk} . The results highlight several key trends and implications that can guide strategic decisions in supply chain management. Below are some findings derived from the analysis:

The analysis presented a nonlinear relationship between the λ_{pk} and total costs. While increasing the λ_{pk} generally reduced costs, the reduction rate was not constant. For instance, the cost reduction percentage was first more significant at lower λ_{pk} (0.10–0.50) than at higher factors (1.00–3.00). This suggests that initial subsidy rises have a more substantial impact on cost reduction than subsequent increases. Managers should consider that initial subsidy increases can lead to significant cost savings, but the marginal benefit drops beyond a certain point.

Green production was initially zero until a λ_{pk} of 0.75, after which it rose significantly. This threshold effect indicates that a certain level of subsidy is required to make green production economically viable. Decision-makers

Table 4 Results of solving experimental examples

Example	GAMS				NSGA-II			
	Objective 1	Objective 2	Objective 3	Number of distributors	Objective 1	Objective 2	Objective 3	Number of distributors
1	9,157,180.65	2,217,880.84	0	1	9,157,265.61	2,228,400.79	0	1
2	17,299,923.15	6,136,220.33	0	3	17,300,278.9	6,266,178.24	0	3
3	22,462,120.60	11,004,563.43	55,511	5	22,462,455.8	11,061,254.3	51,929	5
4	28,647,839.65	6,832,694.46	67,231	5	28,647,993	6,877,759.76	63,596	5
5	33,813,132.42	10,372,890.99	78,103	5	33,813,408.6	10,446,190.6	73,961	5
6	40,352,266.62	15,178,115.80	0	5	40,352,681.1	15,286,145.1	0	5
7	51,577,525.79	22,943,590.80	129,426	5	51,578,713	23,133,311.8	119,684	5
8	58,170,295.12	17,462,012.69	137,538	5	58,175,470.3	17,691,156.5	134,368	5
9	153,446,310.00	45,786,177.33	0	5	153,450,299	46,549,012.6	0	5
10	161,771,356.35	43,576,875.53	0	6	161,774,963	44,158,807.4	0	6
11	173,081,731.12	37,145,730.14	163,504	6	173,089,349	37,813,094.1	152,723	6
12	291,815,928.65	46,155,968.39	352,466	7	291,830,241	47,186,405.9	325,751	7
13	307,806,621.06	52,878,971.54	0	7	307,818,895	53,439,733	0	7
14	339,698,563.52	41,807,624.30	199,994	7	339,723,598	42,091,240.6	182,489	7
15	358,411,720.12	49,062,474.73	0	7	358,482,465	50,653,804.8	0	7
16	552,431,019.62	51,114,154.68	214,206	7	552,466,995	52,876,466.7	206,302	7
17	649,286,178.27	104,137,988.72	820,359	6	649,307,149	105,759,902	812,985	6
18	681,428,008.68	106,858,329.89	265,544	7	681,439,071	108,939,807	261,242	7
19	712,360,368.66	133,704,133.38	547,387	7	712,428,959	137,774,865	537,004	7
20	746,356,146.69	105,364,497.36	852,498	5	746,395,593	108,334,048	836,235	5
21	766,382,370.13	80,135,574.24	590,938	7	766,403,817	81,065,598.4	577,195	7
22	821,069,296.09	119,344,188.66	302,798	6	821,144,347	120,552,206	299,213	6
23	871,024,475.73	209,309,469.49	319,134	7	871,046,900	212,461,589	310,555	7
24	961,058,210.35	130,761,136.20	681,299	7	961,114,439	131,834,578	654,441	7
25	996,219,182.05	165,173,668.36	362,488	8	996,407,889	166,562,632	355,454	8
26	1,318,700,169.43	237,791,844.54	352,859	8	1,321,059,279	239,291,473	347,171	8
27	1,381,588,350.28	218,044,419.45	364,651	6	1,381,830,166	219,739,822	360,035	6
28	1,409,245,660.17	205,925,221.91	787,280	7	1,411,105,709	208,572,031	774,935	7
29	1,461,802,925.65	213,870,108.54	814,382	9	1,461,942,720	217,262,640	791,408	9
30	1,476,326,652.41	310,195,735.75	2,090,147.68	6	1,478,464,694	311,511,492	2,047,397	6

Table 5 Computational results, mean of performance criteria, and calculation time

Example	QM (%)	CM (%)	DM (%)	CPLEX Time (s)	NSGA-II Time (s)	Speed-up (%)
21	93.20	0.11	3.59	1 358.3	210.4	84.5
22	74.80	0.14	3.04	1 540.9	250.2	83.8
23	72.20	0.39	3.35	1 672.4	300.5	82.0
24	51.20	1.05	0.62	1 810.5	523.1	71.1
25	56.70	0.41	3.33	1 975.3	412.2	79.1
26	41.70	0.65	2.26	2 150.8	478.9	77.8
27	37.80	0.78	2.91	2 324.5	556.3	76.1
28	74.50	0.32	2.11	2 499.1	673.4	73.0
29	67.10	0.49	2.17	2 668.7	722.0	72.9
30	62.20	0.58	2.85	2 832.6	830.5	70.7

and managers should understand that subsidies need to reach a critical threshold to encourage green production effectively. Incremental increases below this threshold may not be sufficient to drive significant changes. As the λ_{pk} rises, total costs fall, but emissions do not follow a straightforward pattern. At some points, emissions even increase slightly as subsidies grow (e.g., from λ_{pk} 0.75–1.00). It might happen because of shifts in the production mix or transportation strategies that are more cost-effective but less emission-friendly. Managers should balance cost reductions with environmental goals. While increasing subsidies can reduce costs, they must also consider the potential for unintended increases in emissions.

The analysis reveals a step change in emissions and green production around specific λ_{pk} (0.75 and 1.00). These discrete jumps suggest that the system's response to subsidy changes is not smooth, potentially due to fixed thresholds in decision-making processes such as selecting production technologies or distribution strategies. Understanding these step changes can help in planning for transitions. When passing these thresholds, companies can prepare for significant shifts in operations and inventory.

Beyond the λ_{pk} of 1.25, the cost reduction percentage declines. This saturation effect indicates a limit on how much subsidies can drive down costs. Further increases in subsidies have a progressively lower influence on cost savings. Managers should recognize that additional increases may yield diminishing returns after reaching a certain subsidy level. With higher λ_{pk} factors, the model likely incentivizes using alternative fuels and hybrid vehicles due to the associated subsidies, which can alter transportation costs and emissions profiles, suggesting that investments in greener vehicle technologies might be justified as λ_{pk} increase (Fig. 2).

As the λ_{pk} rises, total costs falls, but emissions exhibit a non-monotonic pattern. At some points, emissions even rise slightly as subsidies grow; for example, between λ_{pk} = 0.75 and 1. This takes place because the lower effective

cost of green production, and in some cases, cheaper but higher-emitting transportation modes selected to minimize cost, incentivizes greater overall volume, which can offset per-unit emission gains. Managers should balance cost reductions with environmental goals. While increasing subsidies can reduce costs, they should also consider the potential for unintended increases in emissions.

The analysis shows clear step changes, as seen between λ_{pk} = 0.75 and 1, in both emissions and green production. These discrete jumps suggest that the system's response to subsidy changes is not smooth, potentially due to fixed thresholds in decision-making processes like selecting green technologies or distribution strategies. Understanding these step changes can help in planning for transitions. When passing these thresholds, companies can prepare for significant shifts in operations and inventory.

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Firms should incorporate the non-linear effects of subsidies into their financial planning, recognizing that early-stage increases in λ_{pk} leads to significant cost savings, but marginal benefits taper off beyond a subsidy factor of approximately 1.5. This implies that aggressive initial subsidy uptake is highly beneficial, yet subsequent subsidy hikes should be carefully assessed against their diminishing returns. Further, managers should prepare for threshold-driven operational shifts: the analysis reveals sharp system responses as λ_{pk} crosses key values of 0.75 and 1.00, so

Table 6 Sensitivity analysis of the λ_{pk}

λ_{pk} factor	Total costs	Cost reduction percentage	Emissions	Green production
0.1	1,488,523,057	– 0.826131829	317,882,433.7	0
0.25	1,487,227,631	– 0.738385287	317,882,433.7	0
0.5	1,485,068,589	– 0.59214105	317,882,433.7	0
0.75	1,482,363,454	– 0.408906898	320,671,703.5	420,070
1	1,476,326,652	0	310,195,735.8	2,090,147
1.25	1,467,043,491	0.628801335	309,902,990.3	2,491,168
1.5	1,457,311,185	1.28802573	309,902,990.3	2,491,168
2	1,416,755,182	4.035114478	384,162,960.6	6,972,760
2.25	1,381,965,060	6.391647263	407,387,264.6	8,374,347
2.5	1,345,023,052	8.893939544	407,387,264.6	8,374,347
2.75	1,308,081,045	11.39623182	407,387,264.6	8,374,347
3	1,271,139,037	13.8985241	407,387,264.6	8,374,347
4	1,123,371,005	23.90769323	407,387,264.6	8,374,347
5	975,602,973.9	33.91686235	407,387,264.6	8,374,347
6	827,834,942.4	43.92603147	407,387,264.6	8,374,347
7	680,066,911	53.93520059	407,387,264.6	8,374,347
8	532,298,879.6	63.94436972	407,387,264.6	8,374,347
9	384,530,848.1	73.95353884	407,387,264.6	8,374,347
10	236,762,816.7	83.96270796	407,387,264.6	8,374,347
11	88,953,570.54	93.97466879	412,011,553.8	8,786,494

inventory strategies, technology upgrades, and workforce training should be coordinated with these subsidy milestones to avoid bottlenecks or underutilized capacity.

At the same time, cost and environmental objectives can conflict when lower effective green production costs result in higher total output or trigger transport choices that heighten emissions; therefore, firms should implement complementary emission controls or optimize logistics, such as route planning or load consolidation, to ensure that cost reductions do not inadvertently compromise environmental performance. As subsidies continue to grow, the model increasingly favors hybrid and alternative-fuel vehicles; companies should thus invest in these green technologies judiciously, timing fleet upgrades to coincide with subsidy availability for maximizing environmental benefit without overspending. Finally, since government subsidy schemes can change rapidly, organizations should closely monitor policy evolution and maintain operational agility, through flexible sourcing contracts, scalable production lines, or modular logistics networks, to capitalize on favorable subsidy adjustments and mitigate risks from sudden policy shifts.

Sensitivity Analysis on γ_v : Insights and Managerial Implications

The sensitivity analysis on the gamma factor (γ), which represents the per-unit subsidy for deploying greener vehicles, reveals a complex relationship between subsidy levels and supply chain costs, as can be observed in Fig. 3. Despite a clear nonlinear decline in total costs as γ rises, both total emissions and green production volumes remain essentially unchanged. This occurs because vehicle subsidies change the transport cost structure without affecting the production mix; managers simply substitute subsidized vehicle types for the same shipment volumes, leaving overall emissions and green-product ratios stable.

Initially, modest subsidies ($\gamma < 10$) yield marginal cost savings ($< 0.1\%$), but once γ crosses approximately 100, cost reductions accelerate, peaking at around a 3% reduction when $\gamma \approx 1000$. This threshold effect indicates that moderate vehicle subsidies alone may be insufficient to justify the higher capital or operating costs of green fleets; however, substantial incentives drive widespread adoption of alternative fuel vehicles and thus deeper cost efficiencies.

This finding is novel compared to existing literature, where most multi-objective supply chain studies focus on production or carbon tax levers, since it isolates the impact

of vehicle-specific subsidies and indicates a decoupling of financial and environmental outcomes. In other words, it challenges the expectation that greener vehicle incentives will automatically deliver emission benefits; instead, it reveals a clear divergence between cost and emission responses under γ policies.

From a policy perspective, our results suggest that γ should reach at least the order of 100 to result in cost efficiencies without unintended operational distortions. Policymakers should, therefore, optimize and adjust vehicle subsidies to exceed this turning point if the primary goal is to lower logistic costs, and should couple such subsidies with direct emission-reduction mandates or production incentives to secure environmental gains.

Sensitivity Analysis on θ : Insights and Managerial Implications

Sensitivity analysis on the excess cost parameter (θ_{pk}) for green production processes is conducted to give insights into the relationship among objective functions. At very low values of θ , close to zero, total costs are notably reduced, as the minimal green premium promotes full adoption of environmentally friendly processes without significant cost impact. As θ_{pk} increases toward 0.75, we observe a pivotal cost inflection—total costs plateau and emissions fall most sharply, marking an optimal balance where the environmental gains per unit cost are maximized. As θ_{pk} grows beyond 2.0, costs rise sharply, demonstrating the limits of cost absorption for green production. Significant reductions in emissions are observed even when θ_{pk} is relatively high. For instance, at $\theta_{pk} = 0.5$, emissions drop to 71 million units, suggesting that initial growths of green production do not compromise

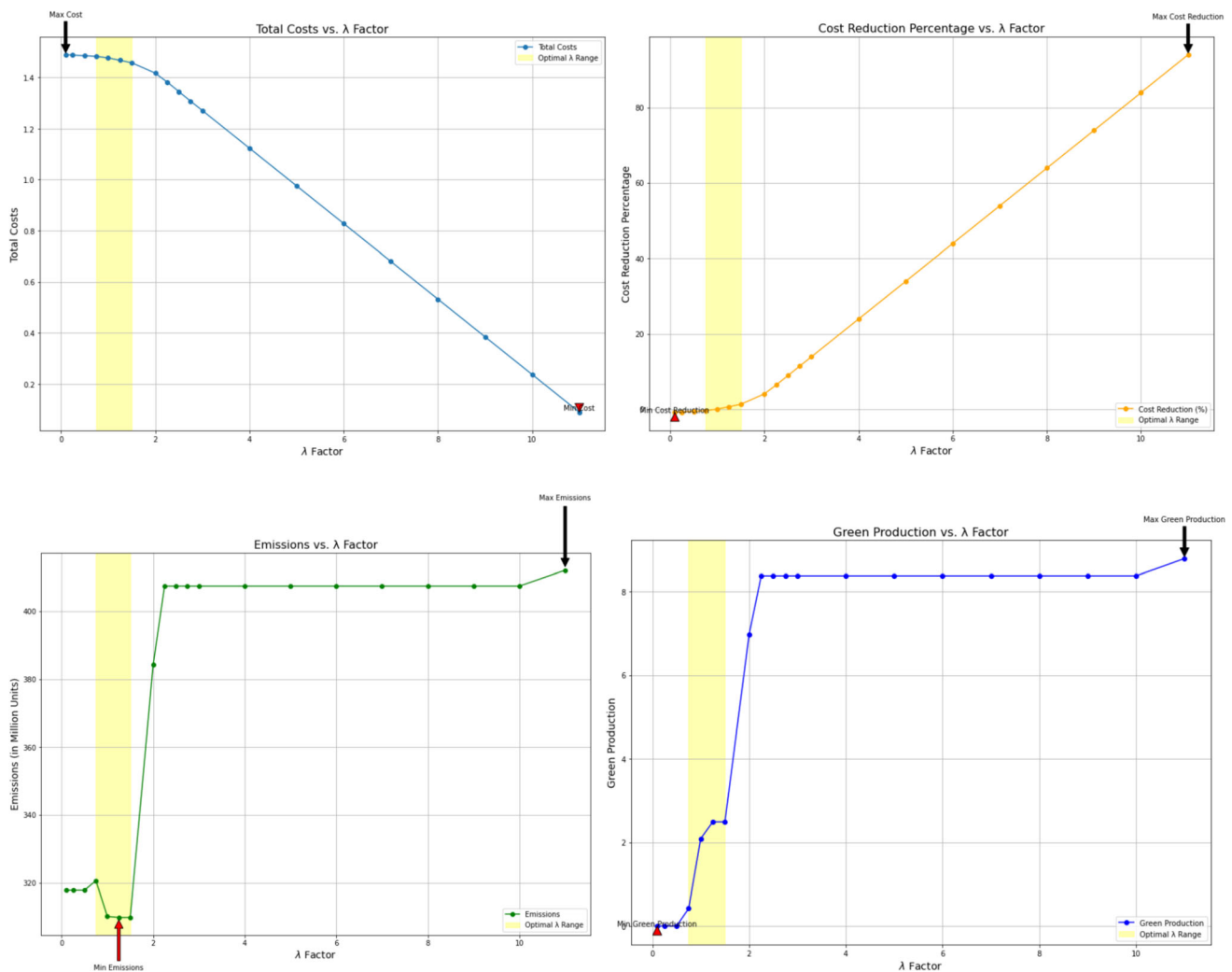


Fig. 2 Sensitivity of costs, emissions, and green production against λ



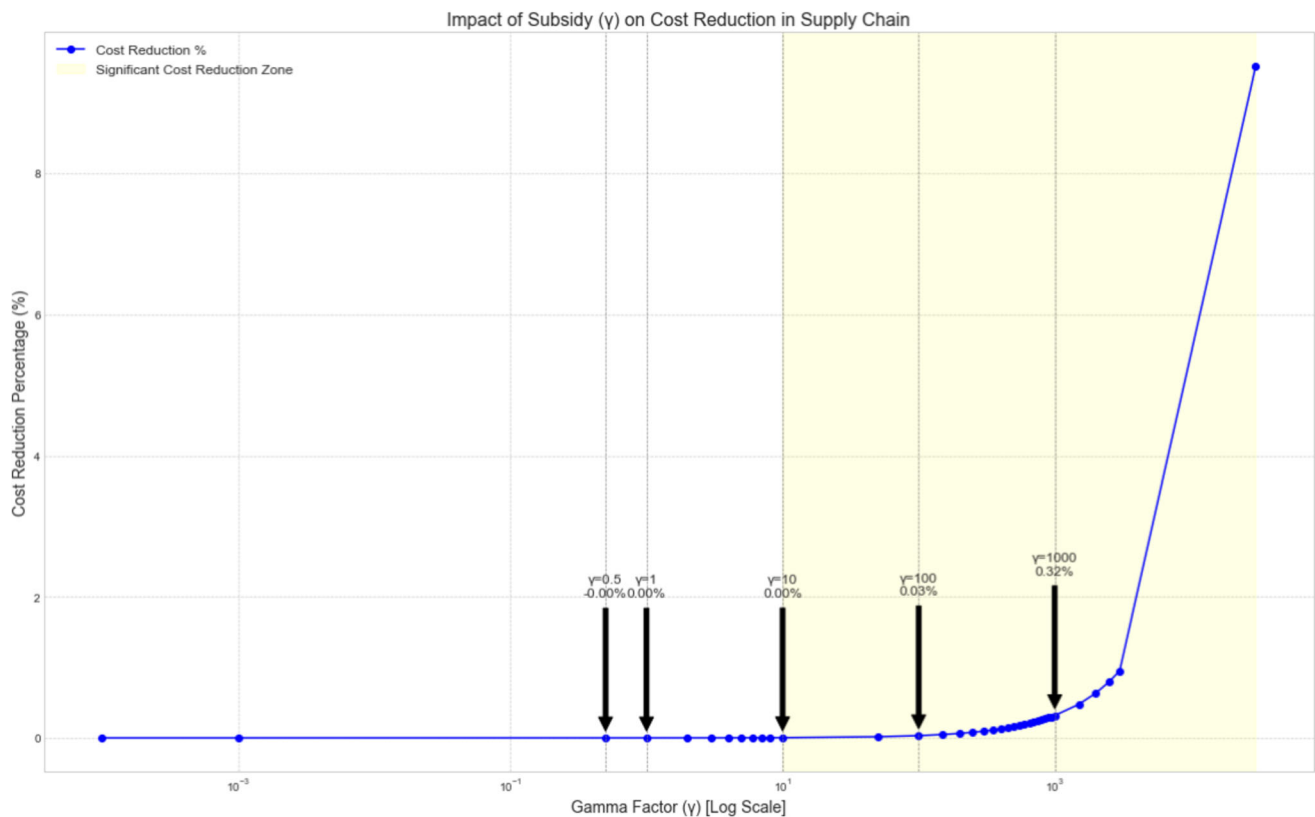


Fig. 3 Sensitivity of costs against γ

environmental benefits. Note that as θ_{pk} increases, the quantity of green production drops significantly.

These insights translate into four key managerial considerations. First, subsidy schemes should be adjusted to offset green premiums up to $\theta_{pk} \approx 0.75$, ensuring cost-competitive green production. Secondly, firms should define cost absorption thresholds, recognizing $\theta_{pk} = 0.75$ as the transition point for switching between green and traditional processes to avoid abrupt shifts in operations. Thirdly, investments in process innovation should target lowering θ_{pk} itself, through R&D or process optimization, to maintain green production viability. Finally, collaborative partnerships, such as public private R&D consortia, can spread the burden of lowering green premium costs and accelerate technology adoption. These strategies help sustain both financial performance and environmental objectives as production costs evolve.

Figure 4 presents an analysis of how variations in the Theta Factor (θ_{pk}) impact total costs, emissions, and green production in the designed network. At $\theta_{pk} = 0.75$, there is a pivotal shift, where the stabilization in cost reduction coincides with a significant decline in emissions, pinpointing an optimal range for operational balance. The strategic annotations highlight critical insights into managing excess costs while maximizing green production,

presenting an innovative framework for sustainable supply chain decisions.

Analysis and Discussion on ε -Constraint

The sensitivity analysis covers a wide range of emission and green production factors, exploring their impact on total costs, actual emissions, and actual green production. It covers 130 scenarios, reflecting various combinations of constraints. Total costs range between 2190 and 195,208,300. According to the results, as emission and green production constraints become more stringent, total costs rise. This is owing to higher expenses associated with advanced green technologies and processes. Actual emissions are also altered within the interval between \$5882 and \$294,132,300. Actual emissions closely follow the imposed constraints, indicating the model's ability to effectively meet environmental targets. Similarly, actual Green Production moved from \$412 to \$20,626,390. The model effectively meets green production targets, revealing flexibility in adapting to higher sustainability goals.

Scenarios with low emission factors (0.001) and high green production factors (50) lead to the highest costs, highlighting the difficulty of achieving aggressive environmental targets. A notable plateau at higher emission

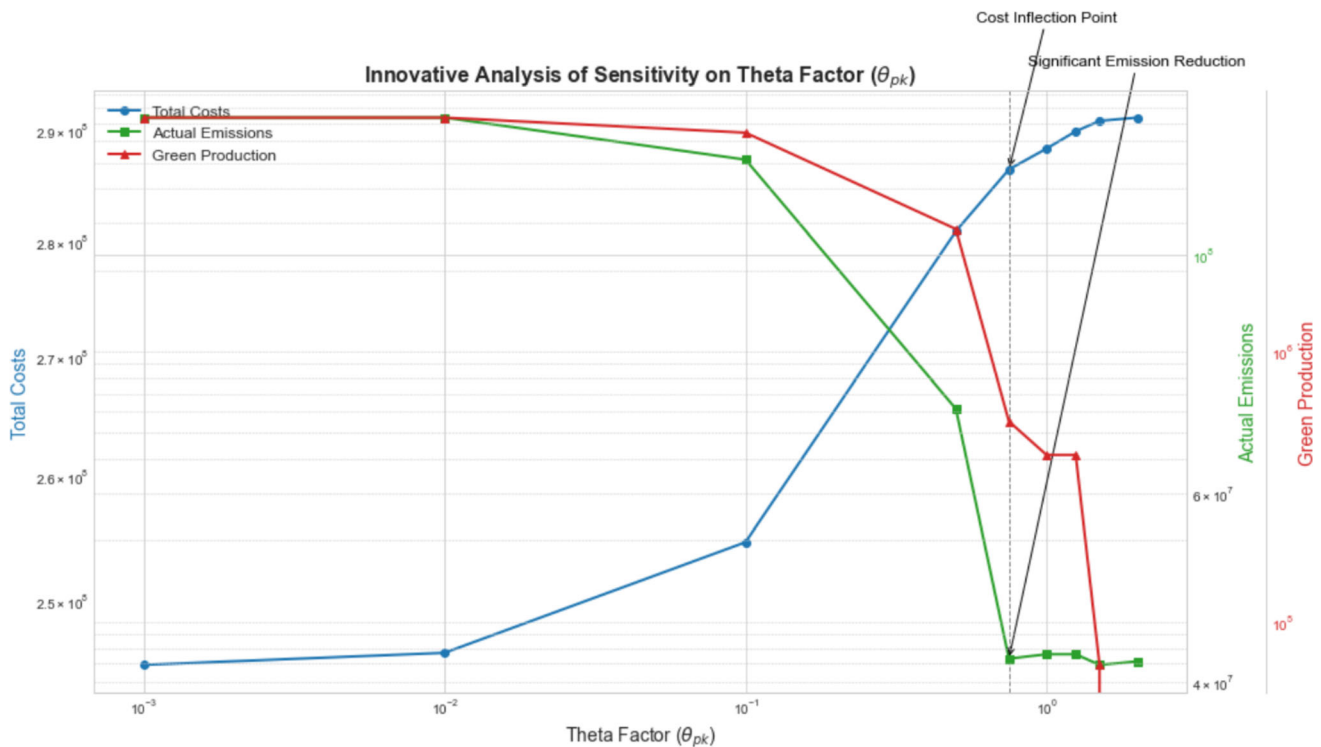


Fig. 4 Sensitivity of costs, emissions, and green production against θ_{pk}

factors shows that once emission constraints are relaxed beyond a certain point, further rises in emission allowances do not significantly lower costs. Further, the rate of cost increase becomes steeper at higher green production factors, emphasizing the exponential growth in costs associated with aggressive green production targets. Scenarios with moderate emission and green production factors indicate a balanced approach, with costs growing moderately compared to baseline values.

The high correlation (0.99) between actual emissions and green production indicates that strategies to raise green production directly impact emissions, indicating a need for planning to balance both objectives. One alternative to balance the situation is to include subsidies for green production and greener vehicles, which help mitigate some cost increases. Nevertheless, the effectiveness of subsidies diminishes as constraints become more stringent. Note that the actual emissions exhibit a linear reduction as emission factors become more stringent.

Green production factors have a minimal direct influence on actual emissions. Thus, while green production processes are environmentally friendly, they primarily affect production sustainability rather than direct emission reduction. At very high green production factors, actual green production levels begin to saturate, demonstrating a potential upper limit to the feasible proportion of green production within the given operational constraints.

- A. Emission factor = 0.001, Green production factor = 50
- A The highest total costs were observed, reflecting significant investments in green technologies.
- B Lowest emissions, fulfilling stringent regulatory requirements.
- C Maximum green production achieved, aligning with sustainability goals.
- B. Emission factor = 1.0, Green production factor = 1.0
- D A moderate rise in total costs, balancing investments in sustainability with operational efficiency.
- E Achievable targets, making it a practical scenario for gradual improvement.
- C. Emission factor = 50, Green production factor = 0.001
- F The lowest increase in total costs, maintaining near-baseline values.
- G Higher emissions represent relaxed environmental standards.
- H Minimal green production, focusing on cost savings rather than sustainability.

Figure 5 indicates the relationship between the emissions factor and total costs, with a trend line fitted. The yellow region in Fig. 5 represents the plateau where further rises of the emissions factor do not significantly lower total costs.



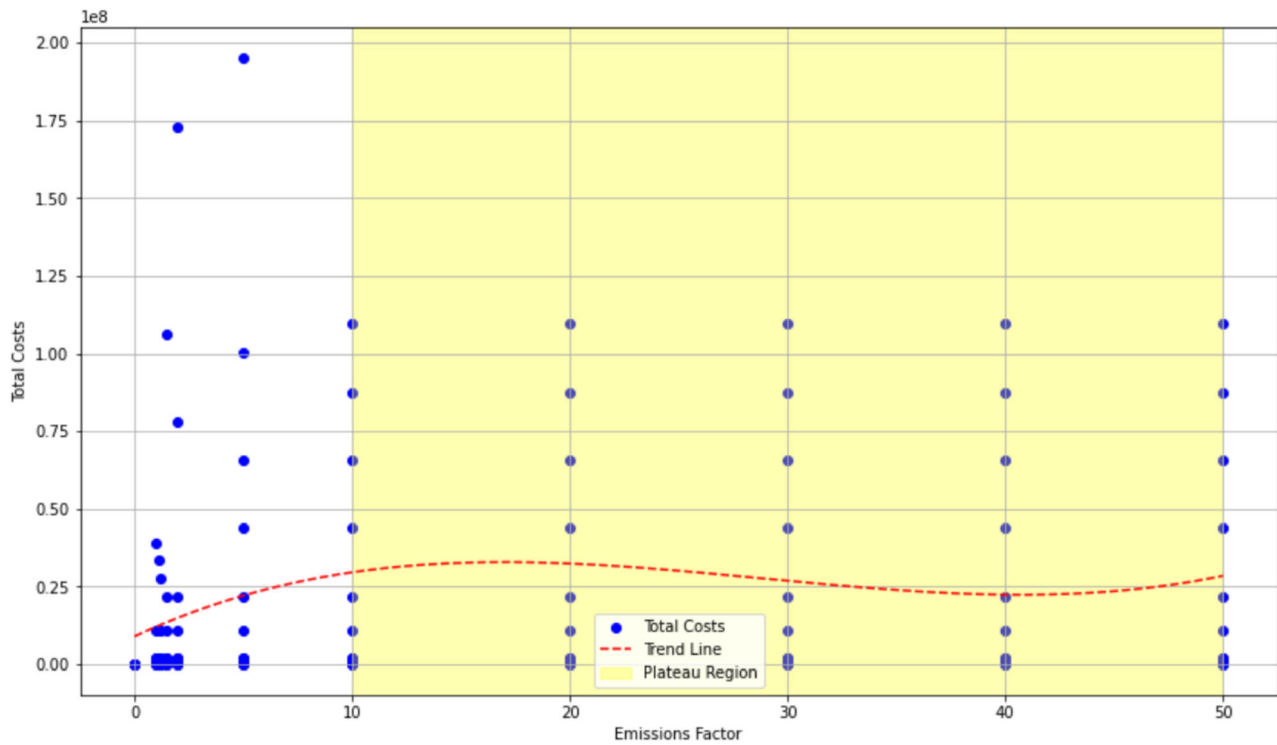


Fig. 5 Scatterplot of total cost vs. emissions

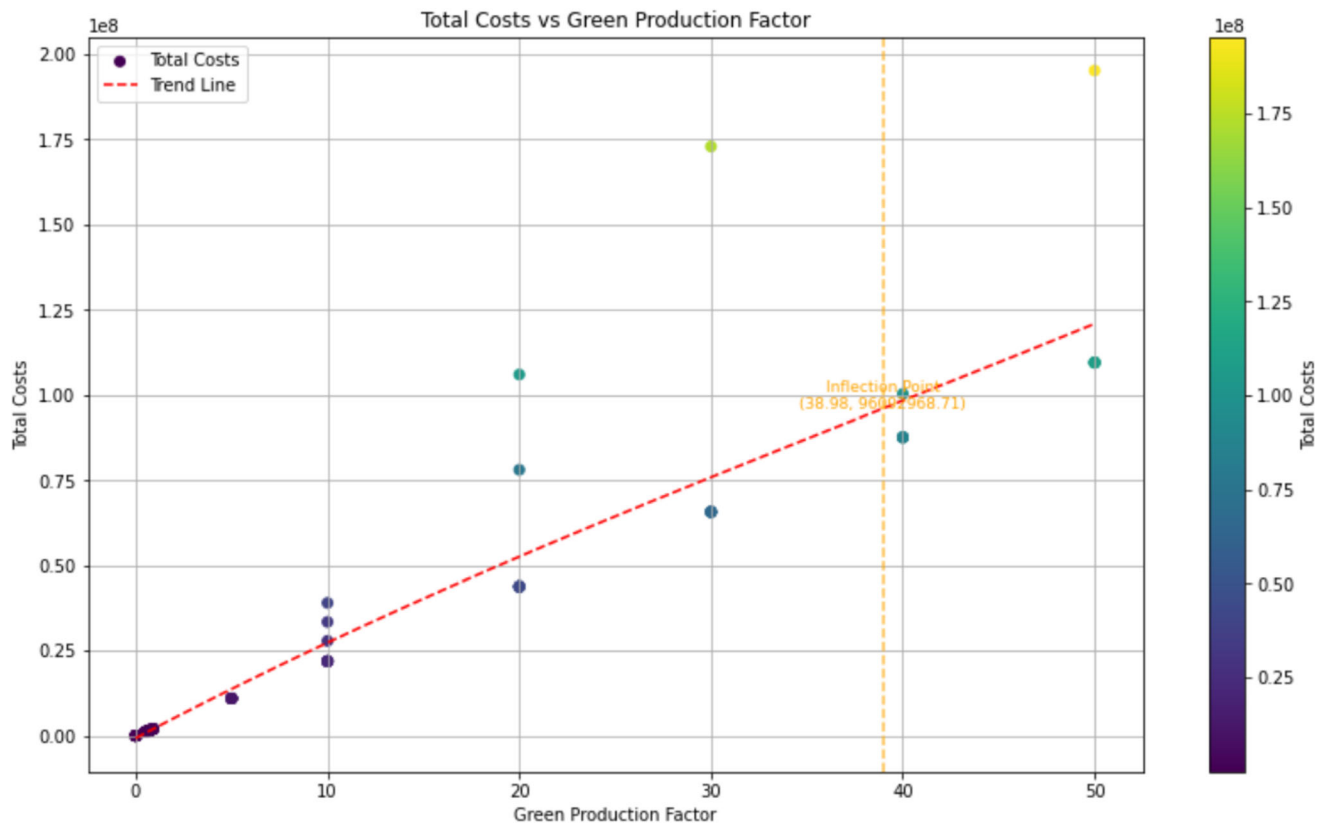


Fig. 6 Scatterplot of total cost vs. green production factor

Figure 6 reveals a clear ascending trend in total costs as green production factors increase, indicating that higher green production targets result in substantial cost increases. The polynomial trend line highlights inflection points, providing critical insights into thresholds where cost increases become more significant. These inflection points are essential for strategic planning, helping managers identify optimal green production targets that balance sustainability and cost efficiency.

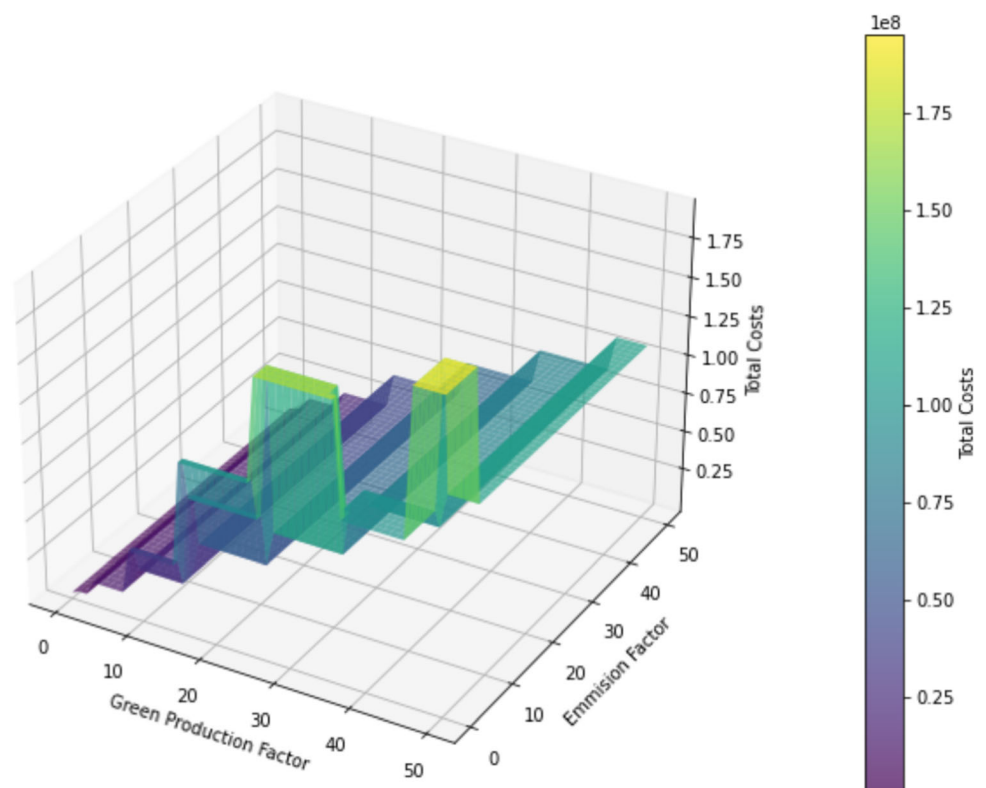
Figure 7 demonstrates the relationship between total costs, green production factors, and emission factors. As green production factors increase, so do total costs generally, representing the higher investment in sustainable practices. Conversely, varying emission factors influence costs nonlinearly, indicating that emission reductions might initially lower costs due to efficiencies but could result in higher expenses with stricter constraints.

Figure 8 displays a negative correlation between green production and total costs, suggesting that higher green production, potentially driven by subsidies and efficient processes, can result in significant cost savings. This counterintuitive finding challenges the traditional notion that sustainability initiatives inherently augment operational costs. Instead, it posits that strategic investments in green technologies not only align with environmental objectives but also boost economic efficiency through diminished operational expenses and subsidies. The

analysis also shows a positive correlation between emissions and total costs, emphasizing the financial burden of green practices. It highlights the importance of integrating emission reduction strategies into operational decision-making to achieve cost-effective sustainability. The substantial reduction in costs associated with lower emissions reflects the potential for innovative policy mechanisms and technological advancements to mitigate the cost impacts of sustainability transitions. This finding provides support for the adoption of green technologies, demonstrating that investments in sustainable practices yield substantial environmental benefits by effectively lowering emissions. It also suggests that green production processes, when optimized and supported by appropriate subsidies, can function as a dual-purpose solution, accomplishing both economic and environmental goals.

Figure 9 depicts a crucial analysis of the relationships between emission limits and actual emissions, indicating significant differences across various scenarios. The emission limit rises stepwise, reflecting the incremental adjustments made to the emission constraints in the sensitivity analysis. This approach tests how the system responds to progressively stricter or more lenient emission limits. The actual emissions reveal a pattern of regular peaks, especially after each step increase in the emission limit. This suggests the system adjusts its operations to maximize efficiency within the permissible emission limits.

Fig. 7 Volumetric diagram of total costs, green production factors, and emission factors



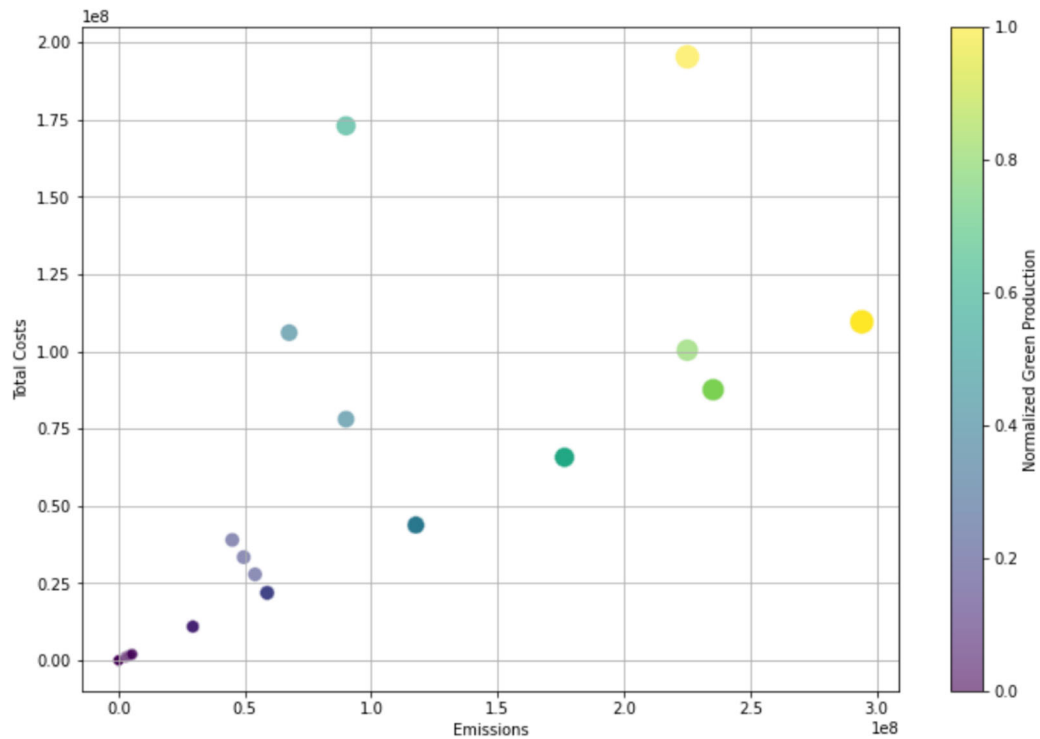


Fig. 8 Scatterplot of total cost vs. normalized green production factor

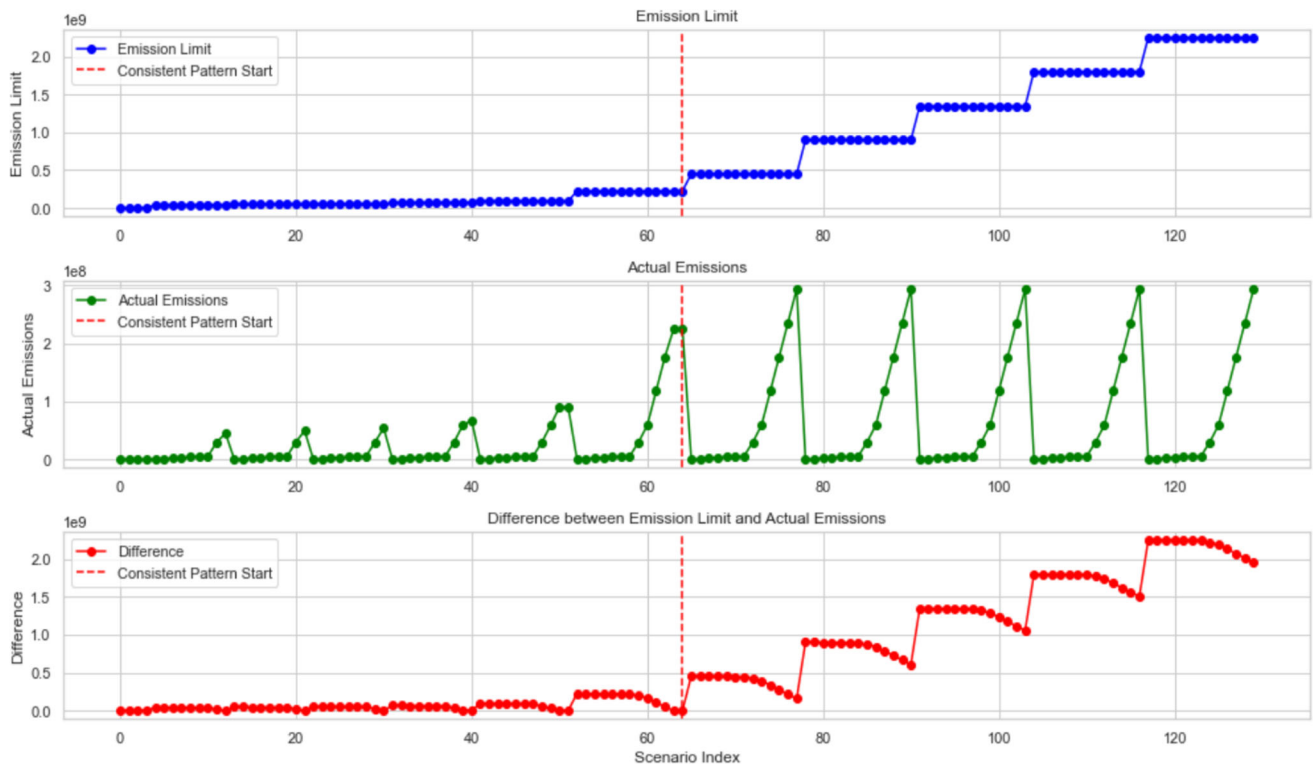


Fig. 9 Emission limits vs. actual emission in different scenarios

The difference between emission limits and actual emissions reveals a consistent pattern of gaps. Initially, the

differences are minimal, representing tight control over emissions. However, as the emission limits increase, the

gaps widen, reflecting the system's ability to lower actual emissions more effectively within higher limits. The patterns in actual emissions suggest that the network adjusts its operations to remain within emission limits. Managers can leverage this adaptive potential to fine-tune production processes and achieve optimal environmental performance.

The consistent pattern in actual emissions starts at scenario #88. At this point, the emission factor is around 1.0, while the green production factor is 0.6. This indicates that when the emission factor reaches 1.0 and the green production factor is set at 0.6, the system stabilizes, maintaining a consistent level of actual emissions. The results indicate that maintaining an emission factor of 1.0 helps in complying with environmental regulations while still operating efficiently. Thus, there is no need to put more emphasis on green factors and impose more costs on the network since the green performance never changes. Decision-makers should target specific emission and green production factor levels to achieve a stable, efficient, and compliant production environment. It, thus, follows that tightening the ε -constraint beyond the identified thresholds leads to only marginal environmental gains while substantially elevating supply chain costs.

Our analysis identifies three key actionable recommendations for policymakers and supply chain managers. Firstly, subsidies should be targeted to critical threshold levels, specifically, green-production subsidies (λ_{pk}) between 0.75 and 1.5 and vehicle subsidies (γ) above 100, to unlock the largest cost savings as well as emission reductions without overspending. Secondly, practitioners should schedule AFV fleet upgrades in tandem with factory retrofit incentives to ensure that transportation and production investments reinforce one another, rather than take place in isolation. Finally, decision-makers should monitor real-time performance metrics, such as green share ratios and marginal emission rates, to adjust subsidy levels, preventing situations where additional incentives yield negligible environmental benefits.

Through focusing on these targeted actions, calibrated subsidy thresholds, synchronized investment planning, and continuous performance monitoring, stakeholders can achieve more consistent and cost-effective sustainability gains.

Flexibility System Management Insights

The developed model translates the principles of flexible systems management into a decision framework for perishable food networks. By allowing managers to modulate two policy-driven levers, production subsidies for green processes and per-trip incentives for alternative fuel vehicles, across three interlinked echelons, the formulation embeds both structural flexibility, through selecting fleet

and facility configurations, and behavioral flexibility, via cost-effective shifts in production and routing decisions. Sensitivity experiments present pronounced threshold effects: a subsidy ratio near 0.75 triggers the first large-scale switch to green production, while vehicle incentives need to exceed approximately 100 currency units before logistics costs fall without worsening emissions. These stepwise transitions demonstrate how the supply chain can reconfigure itself in stages as policy signals change, indicating the agility and resilience mechanisms described in the flexible systems management concept, where adaptive rerouting and dynamic priority balancing are viewed as hallmarks of viable operations under volatility.

Beyond demonstrating agility, the study enriches management of flexible systems by quantifying how policy-induced options reshape the trade-off landscape among cost, emissions, and social welfare. The discovery that emissions may increase once production subsidies pass roughly 1.5, in spite of continual cost savings, exposes a non-monotonic response that pure cost or emission models overlook. Likewise, the finding that vehicle subsidies alone decouple financial and environmental outcomes challenges the widespread assumption that greener fleets automatically deliver cleaner performance. These results extend the field in three ways: they provide numerical evidence that flexibility is governed by identifiable subsidy thresholds rather than smooth gradients, they reveal that structural and behavioral levers interact multiplicatively rather than additively, and they offer actionable guidance for calibrating incentives so that flexibility translates into triple-bottom-line goals. Consequently, this study moves flexible systems management from qualitative prescriptions toward a rigorously tested, policy-aware optimization approach appropriate for complex food logistics.

Conclusion and Future Research Directions

This study presented a comprehensive multiple objective optimization model for a three-tier food supply chain which unifies cost efficiency, emission control, and proportion of green output under government subsidy schemes. By integrating production planning, distribution routing, and retail allocation with financial incentives for green manufacturing and cleaner vehicle use, the framework indicates rich interactions among policy levers, operational expenses, and environmental performance. The analysis specifically revealed that modest subsidy levels can drive substantial reductions in total network cost and carbon emissions until specific thresholds are reached. Beyond these thresholds, marginal gains shrink, and unintended effects can emerge, highlighting the critical importance of targeted subsidy calibration.



The model's adaptability to varying subsidy levels and operational conditions underscores its alignment with the principles of flexible systems management. Experiments indicated that the adapted NSGA II delivers near Pareto optimal front quality when compared to an exact solver, yet it requires a fraction of computational time. This suggests that metaheuristic methods can guide real-time decision support even as supply chain complexity grows. Sensitivity analyses unveiled step changes in the adoption of green processes and vehicle fleets as subsidy parameters change, while excess cost factors for green production define clear cutoffs where traditional methods regain preference. These insights equip managers and policymakers with practical guidelines on how to design subsidy schedules, time fleet upgrades, and invest in process innovation to achieve balanced economic and environmental outcomes.

Although the dairy case study anchors our numerical work, the conceptual architecture of the model remains relevant across a wide range of perishable goods networks. Fresh produce, meat, seafood, and beverage supply chains share common features such as limited shelf life, multiple distribution layers, as well as heterogeneous product portfolios. By recalibrating parameters such as spoilage rates, production cost premiums, and vehicle capacities, the same framework can be applied to assess subsidy impacts and fleet modernization strategies in these sectors. This generalizability makes our approach a versatile toolkit for both researchers and practitioners intent on harmonizing public incentives with private investment decisions.

Furthermore, the model aligns with recent developments in flexible systems management by providing a structured yet adaptable approach to managing uncertainties in supply chains, such as policy shifts or market volatility. This capability is critical in environments characterized by frequent disruptions and evolving regulatory landscapes, where rapid adaptation becomes essential for maintaining competitiveness and sustainability. Hence, integrating principles of flexible systems management into supply chain decision-making not only mitigates risks but also leverages uncertainty to create a strategic advantage.

Further extensions can broaden applicability. Incorporating stochastic demand processes captures market volatility and enables robust network design under uncertainty. Introducing dynamic pricing or willingness-to-pay formulations could reflect consumer response to green attributes and enable analysis of profit-driven trade-offs. Incorporating cooperative game frameworks among supply chain partners would provide deeper insights into contract design and equitable profit allocation when multiple stakeholders share green investments and policy benefits.

Regarding the solution methodology, hybrid approaches that blend evolutionary metaheuristics with localized search or machine learning-guided parameter tuning could

further improve convergence speed and solution diversity, enabling real-time adaptation as market conditions evolve. Examining adaptive mechanisms such as population resizing or dynamic mutation rates may help maintain performance even as the problem scale grows beyond the scenarios tested here.

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Data Availability The data used in this research are available upon request.

Declarations

Conflict of Interest The authors declare that they have no conflicts of interest. None of the authors is a member of the Editorial Board or a Guest Editor of the Global Journal of Flexible Systems Management.

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Key Questions

1. How does the integration of government subsidies into supply chain decision-making offer practical insights into the adoption of green logistics practices?
2. In what ways could the model be enhanced to account for dynamic or uncertain factors such as price volatility, demand shifts, or supply chain disruptions?
3. How might stakeholder behavior (e.g., consumer preferences, supplier resistance) influence the effectiveness of subsidies in practice?

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